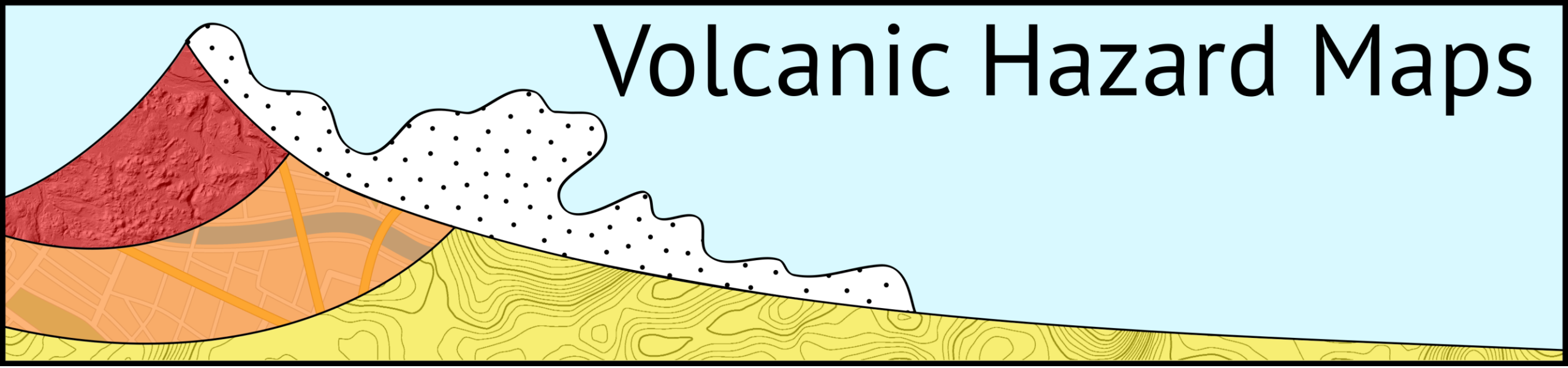


Volcanic Hazard Maps

Sarah Ogburn

IAVCEI

Commission on Volcanic HAZARD & RISK

VDAP Volcano Disaster Assistance Program

THE UNIVERSITY OF AUCKLAND Te Whare Wānanga o Tāmaki Makaurau NEW ZEALAND

THE UNIVERSITY of EDINBURGH

GSJ

Volcanic Hazard Maps

- important communication tool
- are the visual “face” of our knowledge
- form focus of discussions
- for geologists, they are the end product of our work; but they are the beginning of the ***impact***
- There is not a one-size-fits-all approach for constructing a hazard map!



Hazard map of Cotopaxi volcano

Introduction

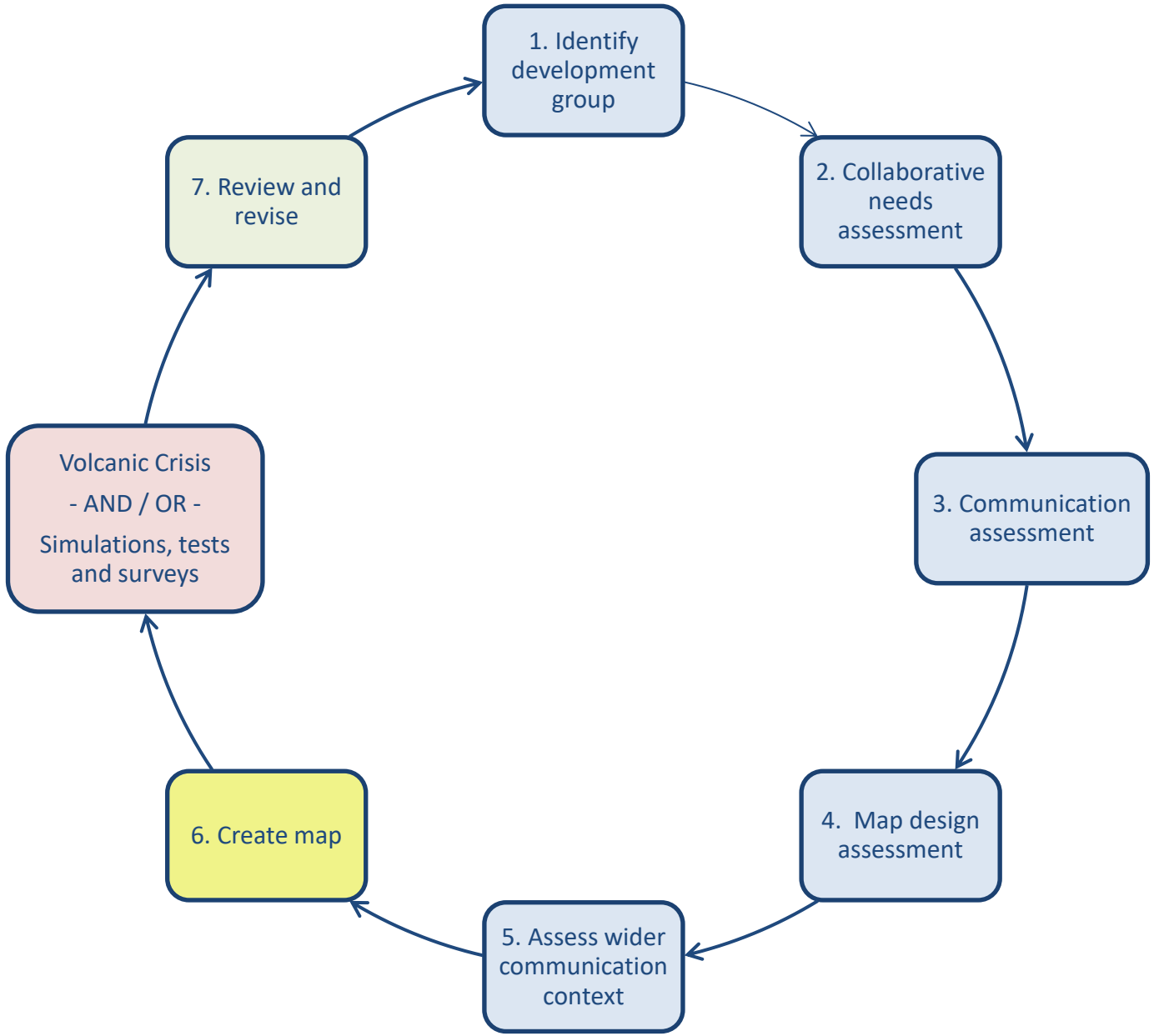
- Ten workshops over 10 years with >250 people attending
- Near completion (later this year) of compiling what was learned from these sessions into best practices book:

The IAVCEI companion to volcanic hazard maps and map making

Jan Lindsay, Eliza Calder, Sarah Ogburn, Daniel Bertin, Heather Wright, Danielle Charlton, Bastian Steinke





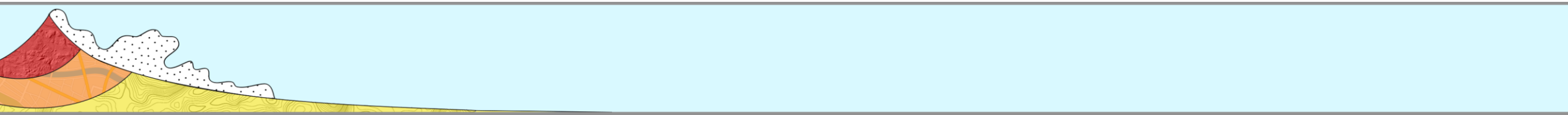


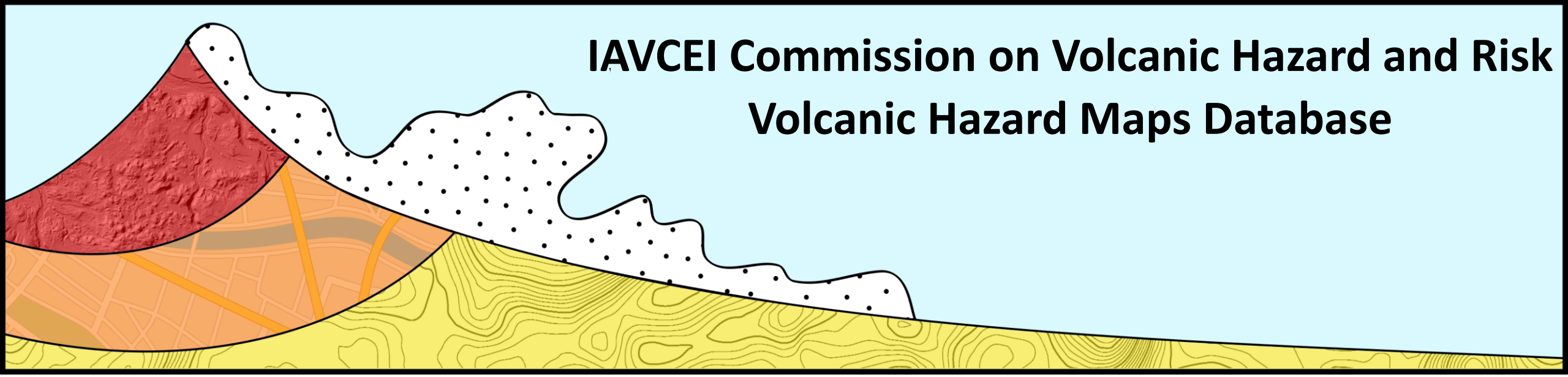
An Idealized Hazard Map Development Process

- Maps need to be fit to purpose and audience
- The information, communication, and design needs of user groups are considered during map development
- The map is evaluated using simulations, tests, or surveys OR during actual volcanic crises
- The map is reviewed and potentially revised according to the evaluation results

However, very few actual hazard maps have been developed using this “ideal” workflow because it requires time, money, personnel, community interest, etc. It’s difficult!

But perhaps we can learn from exploring the diversity of existing hazard maps…

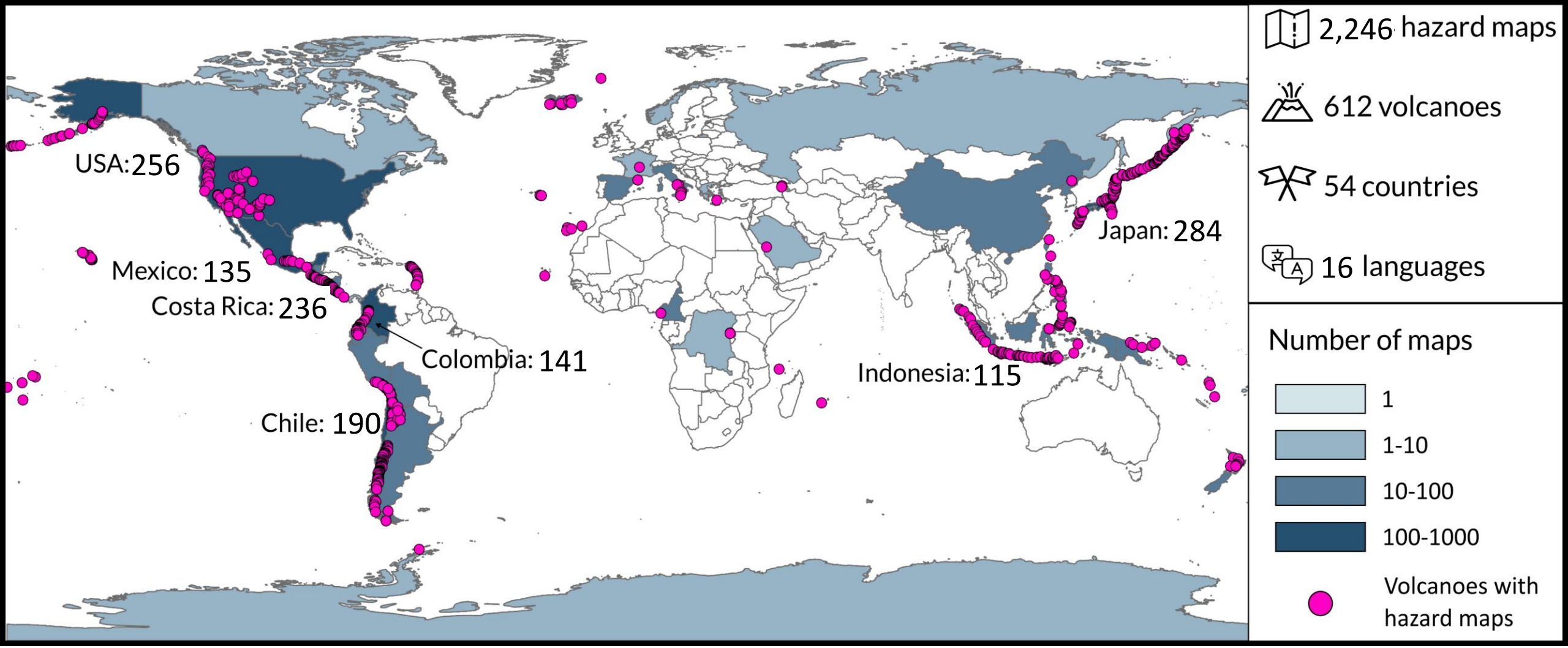


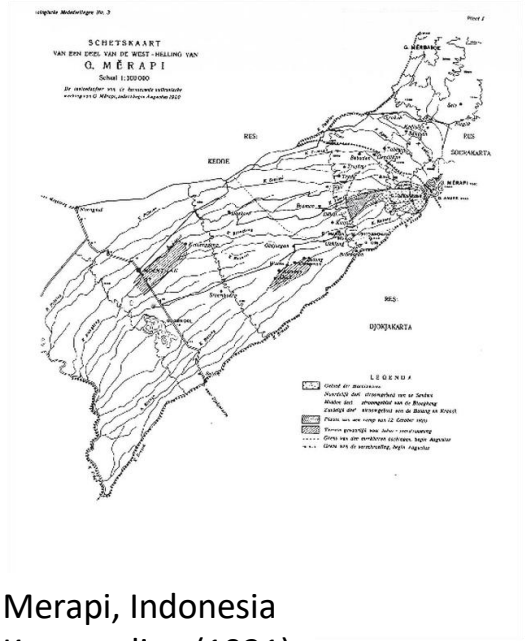


IAVCEI Commission on Volcanic Hazard and Risk
Volcanic Hazard Maps Database

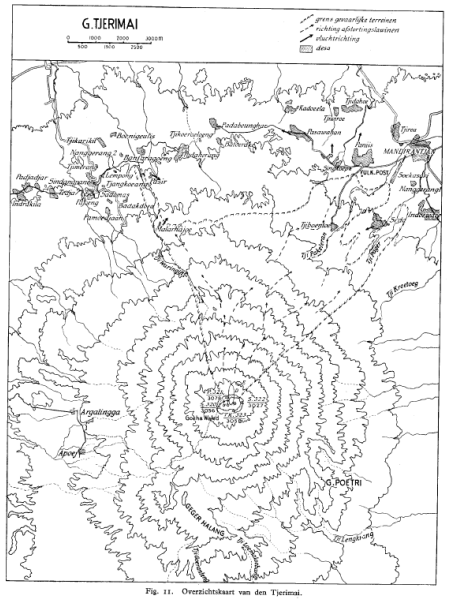
The main purpose of the database and its accompanying website (https://volcanichazardmaps.org/) is to serve as a resource for hazard mappers to explore how common aspects of hazard map development and design have been addressed in different countries, for different hazards, and for different intended purposes and audiences.

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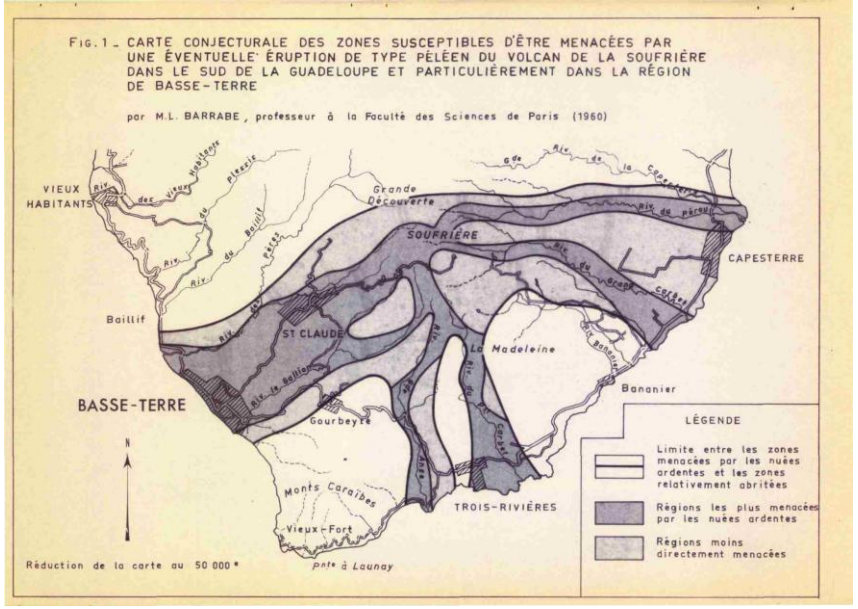

USA: 256
Mexico: 135
Costa Rica: 236
Colombia: 141
Chile: 190
Indonesia: 115
Japan: 284
2,246 hazard maps
612 volcanoes
54 countries
16 languages
Number of maps
1
1-10
10-100
100-1000
Volcanoes with hazard maps



Merapi, Indonesia
Kemmerling (1921)



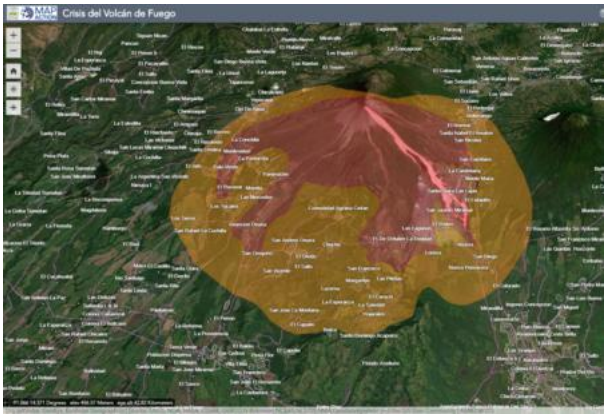
Cereme, Indonesia
Neumann van Padang (1937)



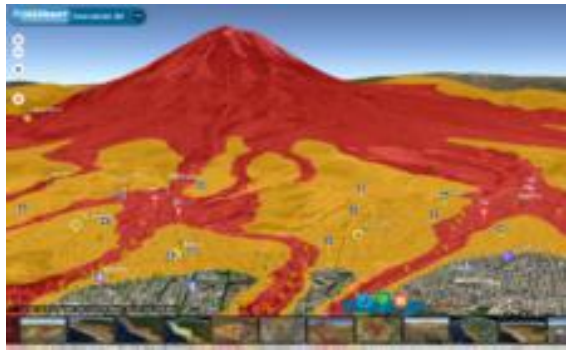
Soufrière Guadeloupe, Guadeloupe [France]
Barrabe (1960)



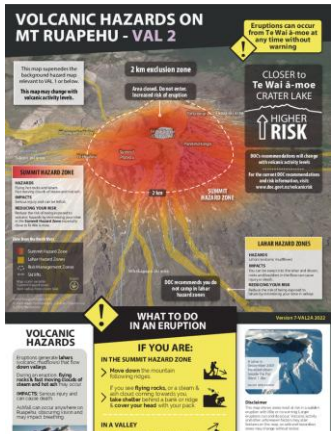
Kamchatka, Russia
Markhinin (1962)



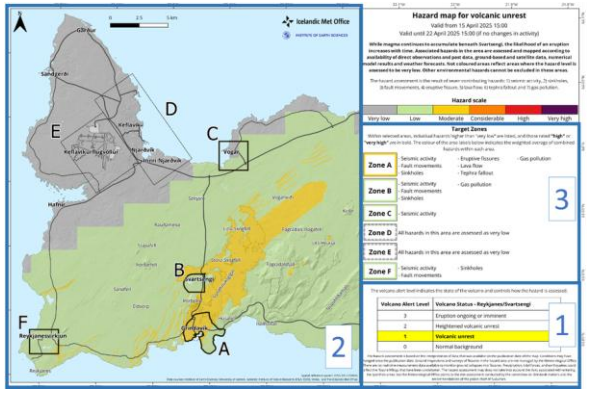
Fuego, Guatemala
INSIVUMEH, Map Action, 2018



El Misti, Peru
INGEMMET, 2019



Ruapehu, NZ
GNS Science, 2022



Svartsengi (Reykjanes), Iceland
Icelandic Meteorological Office, 2025

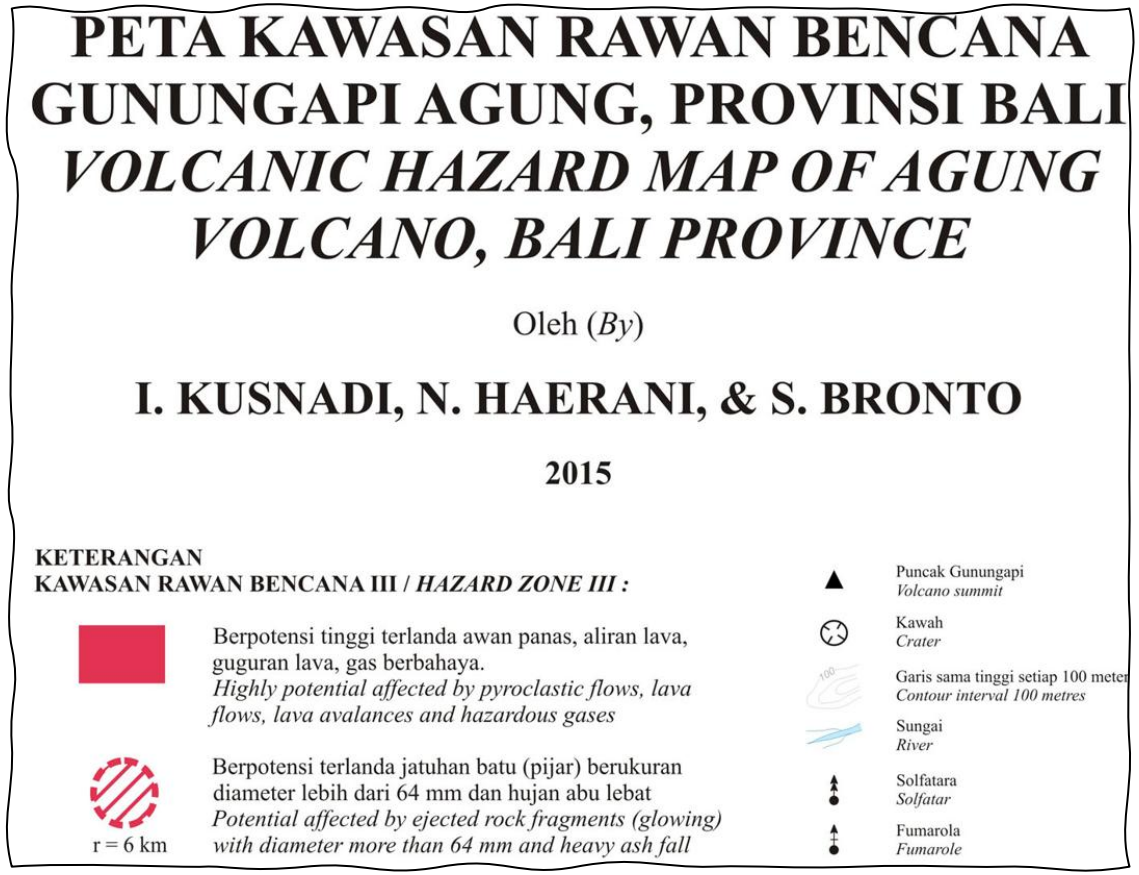
Database contains maps dating from 1921-2026

VDAP USGS

Language

16 different languages represented in the database

- About 6% of maps display text in multiple languages.
- Other maps have multiple versions in different languages.



PETA KAWASAN RAWAN BENCANA GUNUNGAPI AGUNG, PROVINSI BALI
VOLCANIC HAZARD MAP OF AGUNG VOLCANO, BALI PROVINCE

Oleh (*By*)

I. KUSNADI, N. HAERANI, & S. BRONTO

2015

KETERANGAN
KAWASAN RAWAN BENCANA III / *HAZARD ZONE III :*

Berpotensi tinggi terlanda awan panas, aliran lava, guguran lava, gas berbahaya.
Highly potential affected by pyroclastic flows, lava flows, lava avalances and hazardous gases

Berpotensi terlanda jatuhan batu (pijar) berukuran diameter lebih dari 64 mm dan hujan abu lebat
Potential affected by ejected rock fragments (glowing) with diameter more than 64 mm and heavy ash fall

r = 6 km

Puncak Gunungapi
Volcano summit

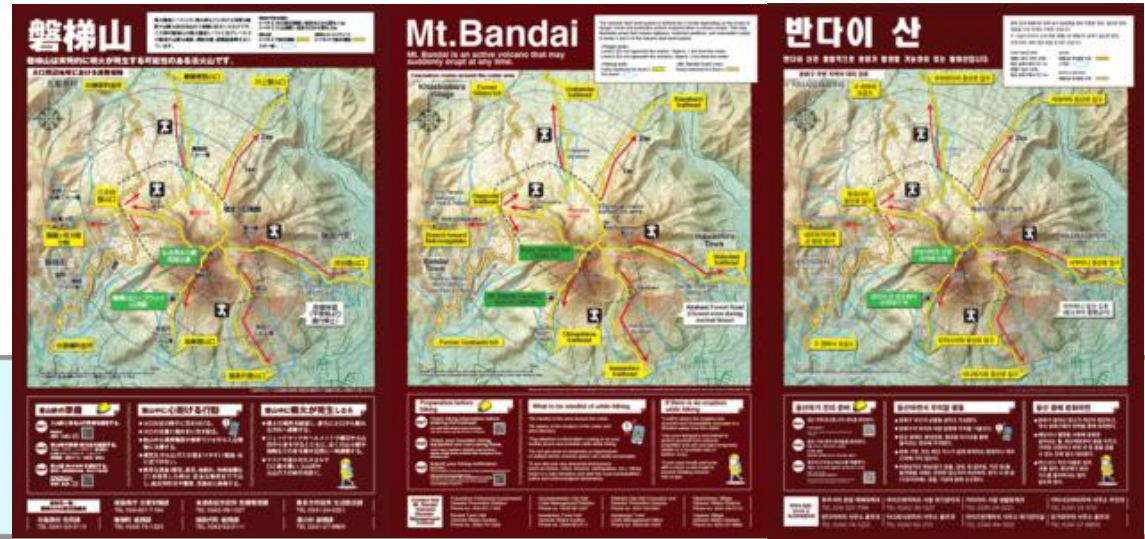
Kawah
Crater

Garis sama tinggi setiap 100 meter
Contour interval 100 metres

Sungai
River

Solfatara
Solfatar

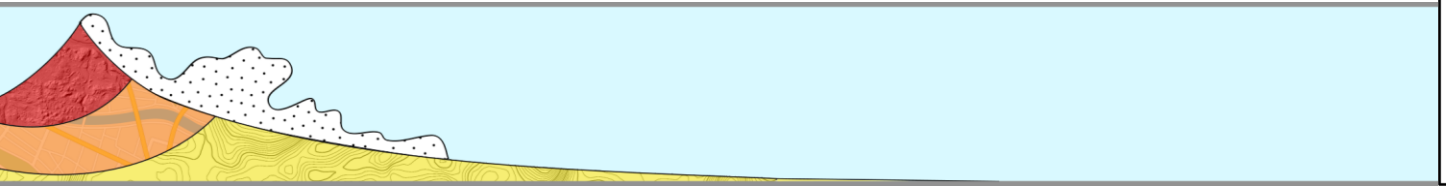
Fumarola
Fumarole

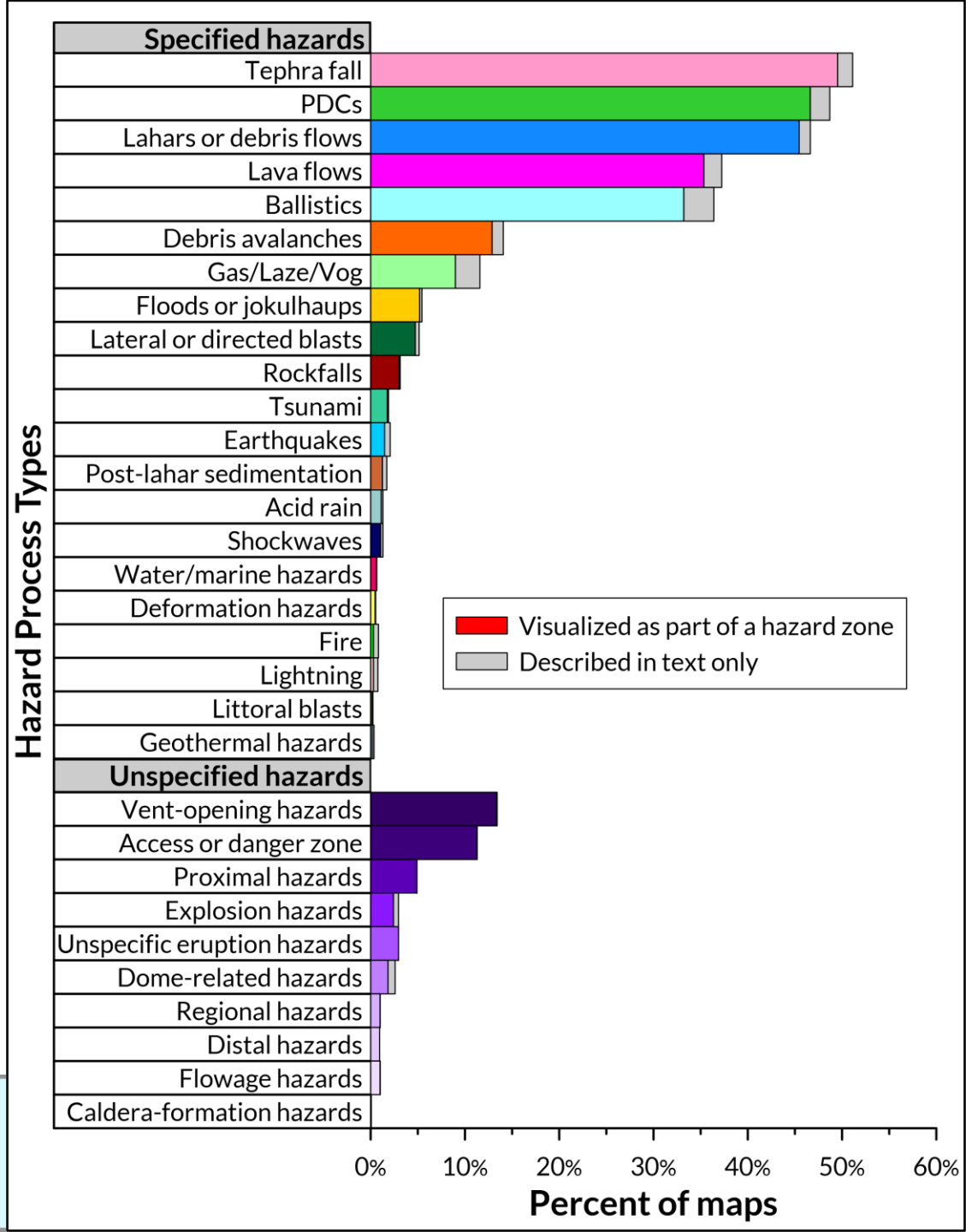




Hazard Processes

- Which hazard processes are portrayed on a map depends on the volcano, but also on the temporal and spatial scale of the map and on the map purpose and audience.
- For example, low-probability, high-impact events (debris avalanches) and longer-term secondary hazards (aggradation) might only be portrayed on maps with long-term temporal scales.
- Tephra fall (50%), PDCs (incl. surges and ash-cloud surges, 44%), lahars (44%), lava flows (34%), and ballistics (31%) are the most commonly portrayed hazards.
- Some hazards might only be described in the text, rather than visualized in a hazard zone (e.g., gas hazards).



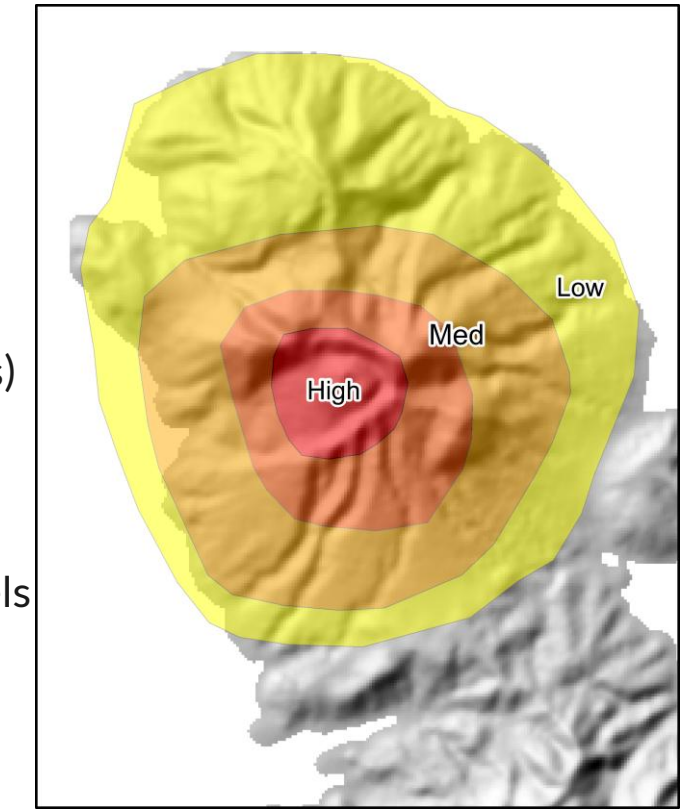


Hazard Zone Presentation

Hazard maps can be classified into two main types according to whether the map is *hazard level-focused* with hazard processes integrated into combined zones or *hazard process-focused* with different zones for each hazard process.

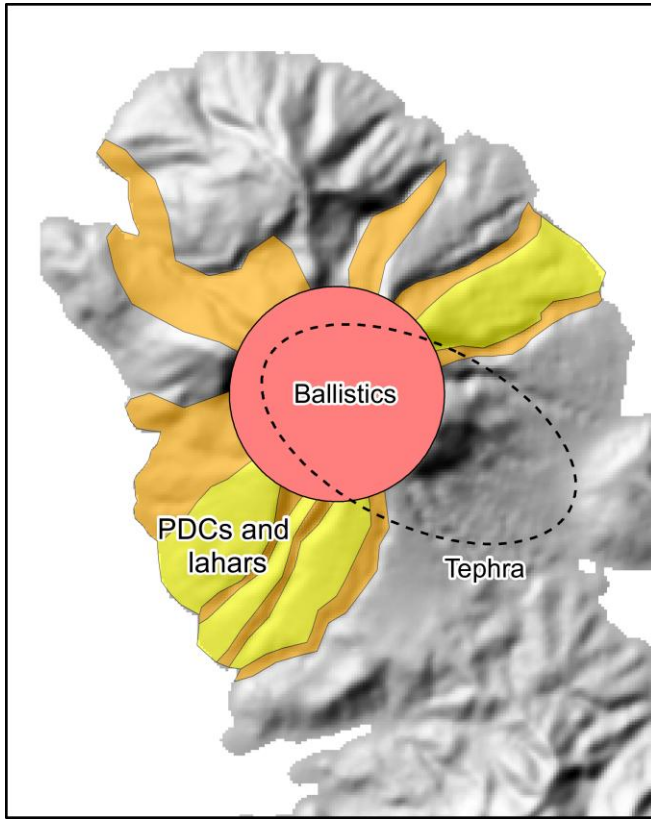
Hazard level-focused

- combine all or most hazard processes into integrated hazard level zones
- Zones are often defined by relative level of hazard (e.g. high-med-low hazard zones)
- The relative hazard level is easy to understand; especially effective for communicating hazard levels to the public.
- Basis for zones not very transparent.



Hazard process-focused

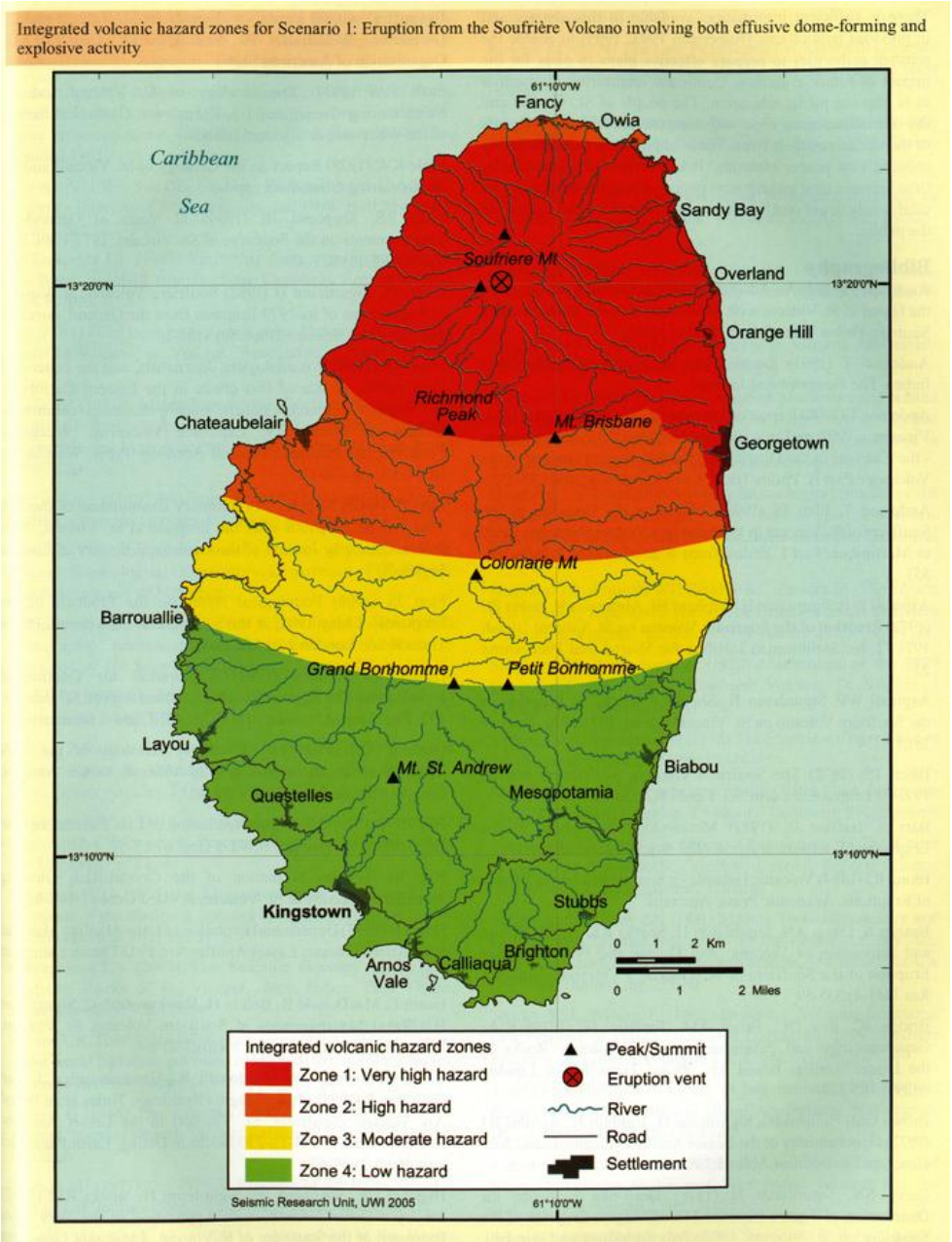
- different zones for each hazard process under consideration.
- Zones are often defined by the hazard process type (e.g lahar hazard zone, tephra hazard zone)
- Geologic reasoning behind the hazard zones is easily perceived, but relative hazard level may be more difficult to understand
- Especially useful for land-use planning, because different hazards have different impacts



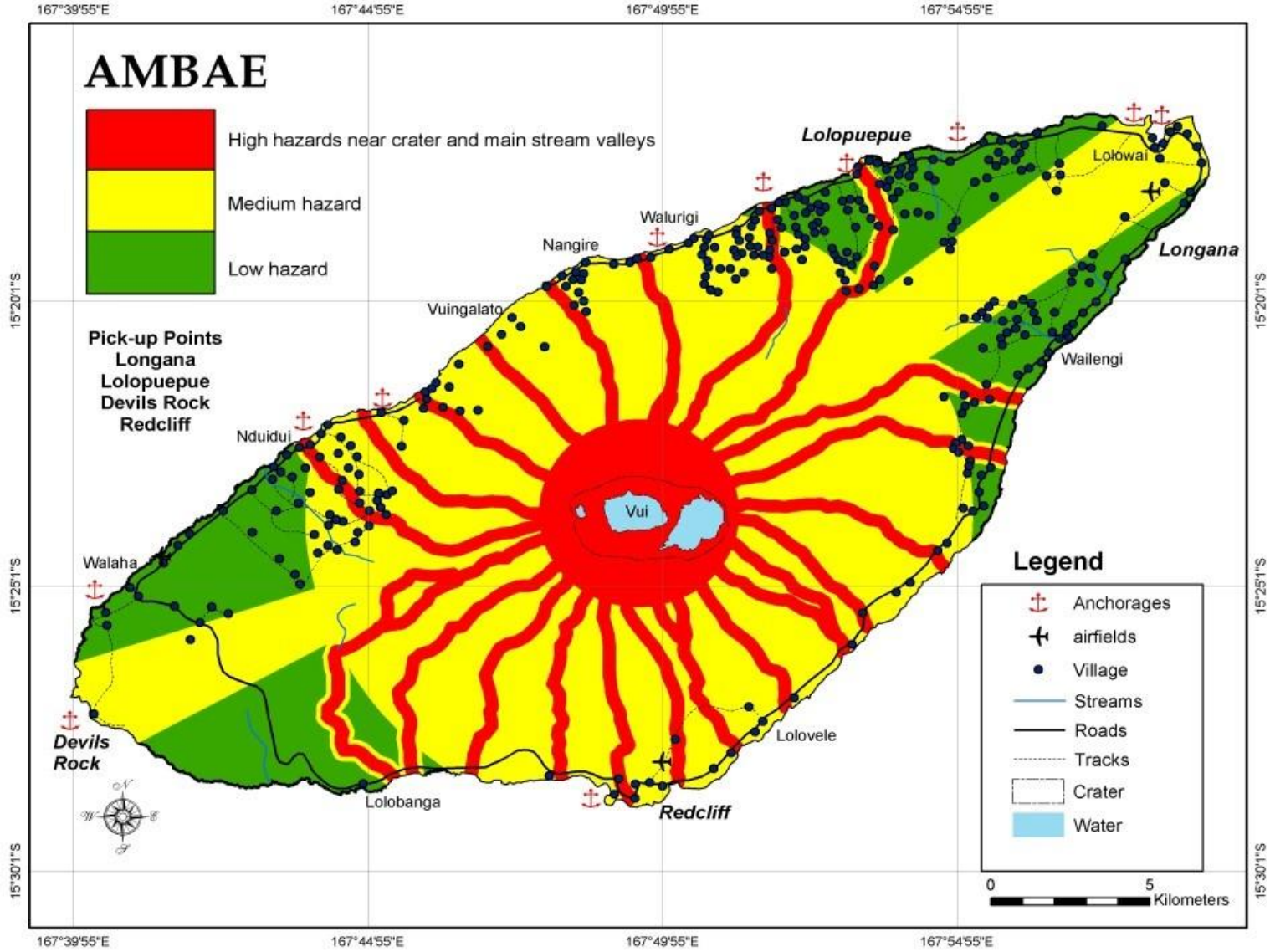


VDAP USGS

Hazard level-focused (integrated)

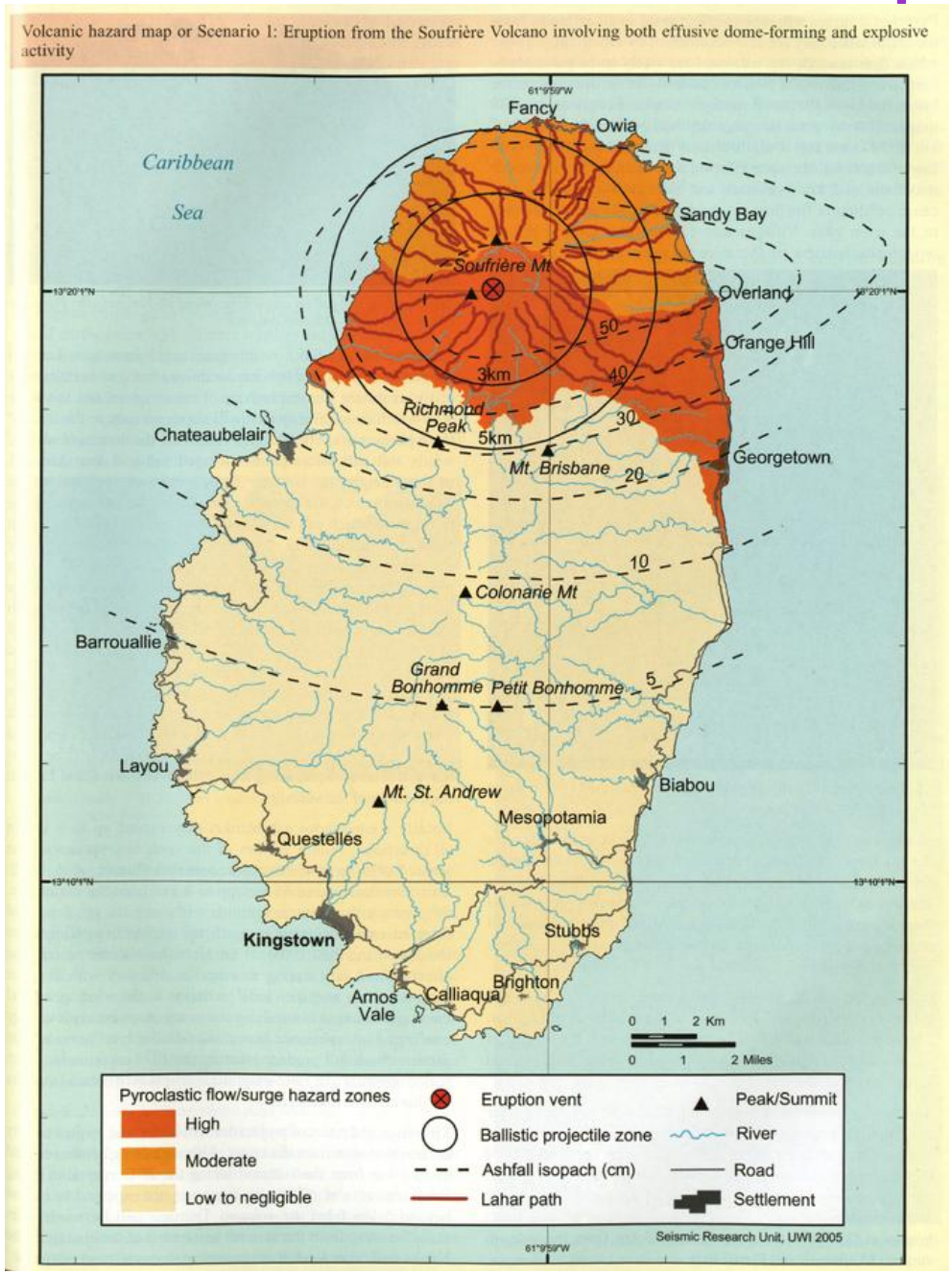


Soufrière St. Vincent (Robertson, 2005)

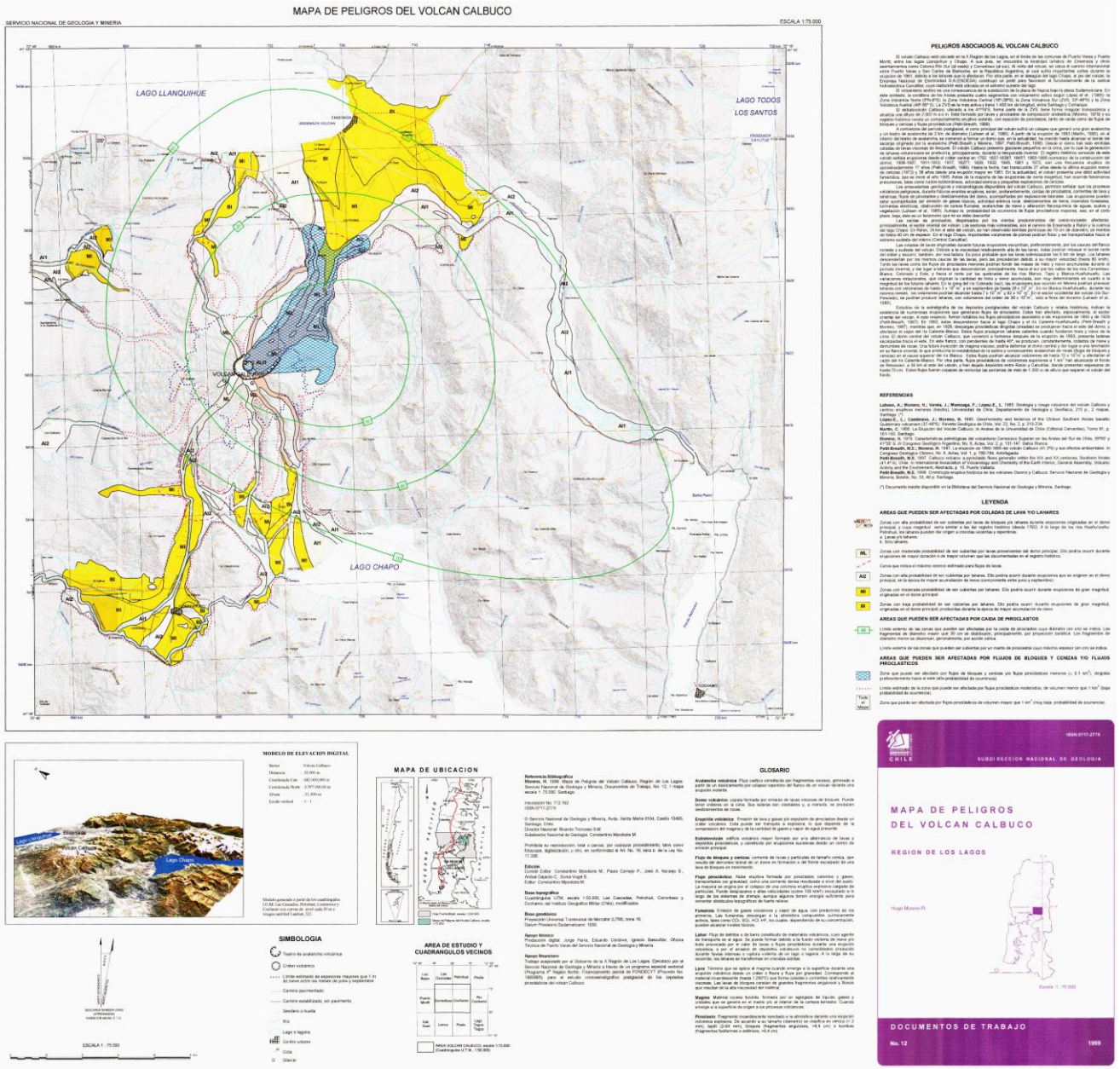


Ambae, Vanuatu, 2001 (Cronin et al. 2004)

Hazard process-focused (separated)

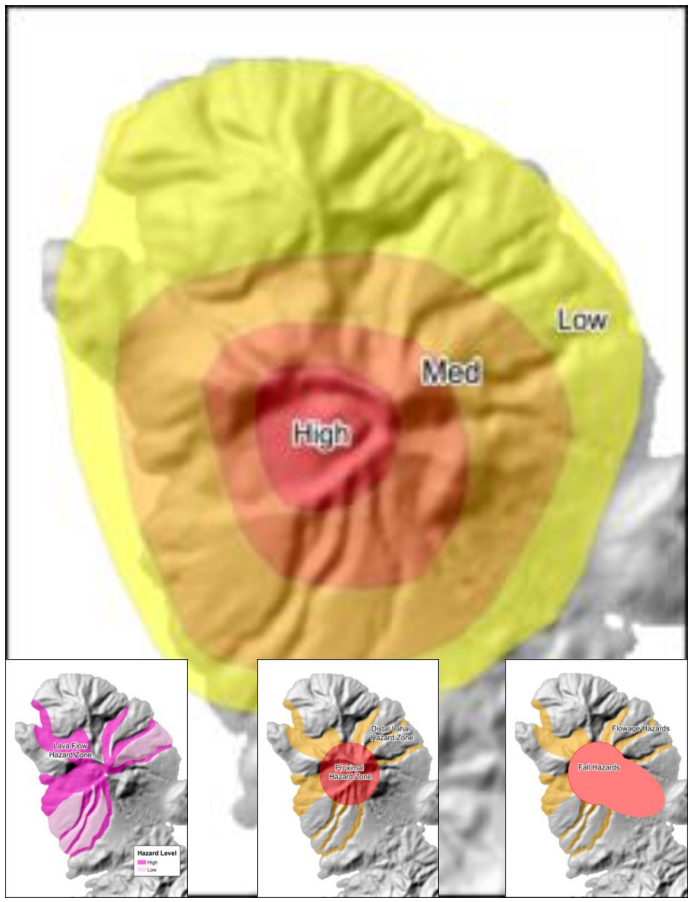


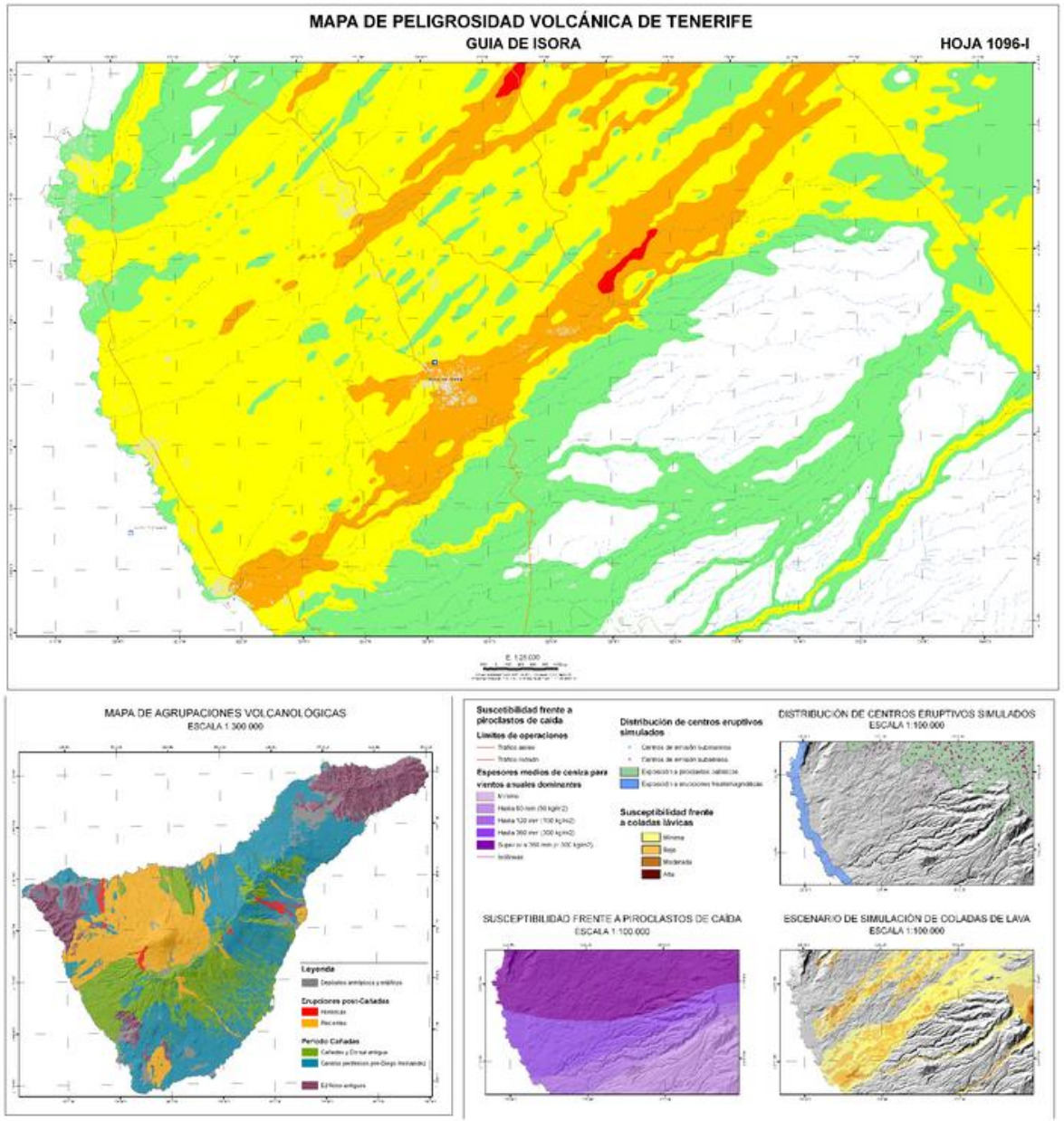
Soufrière St. Vincent (Robertson, 2005)



Calbuco, Chile (Moreno, 1999)

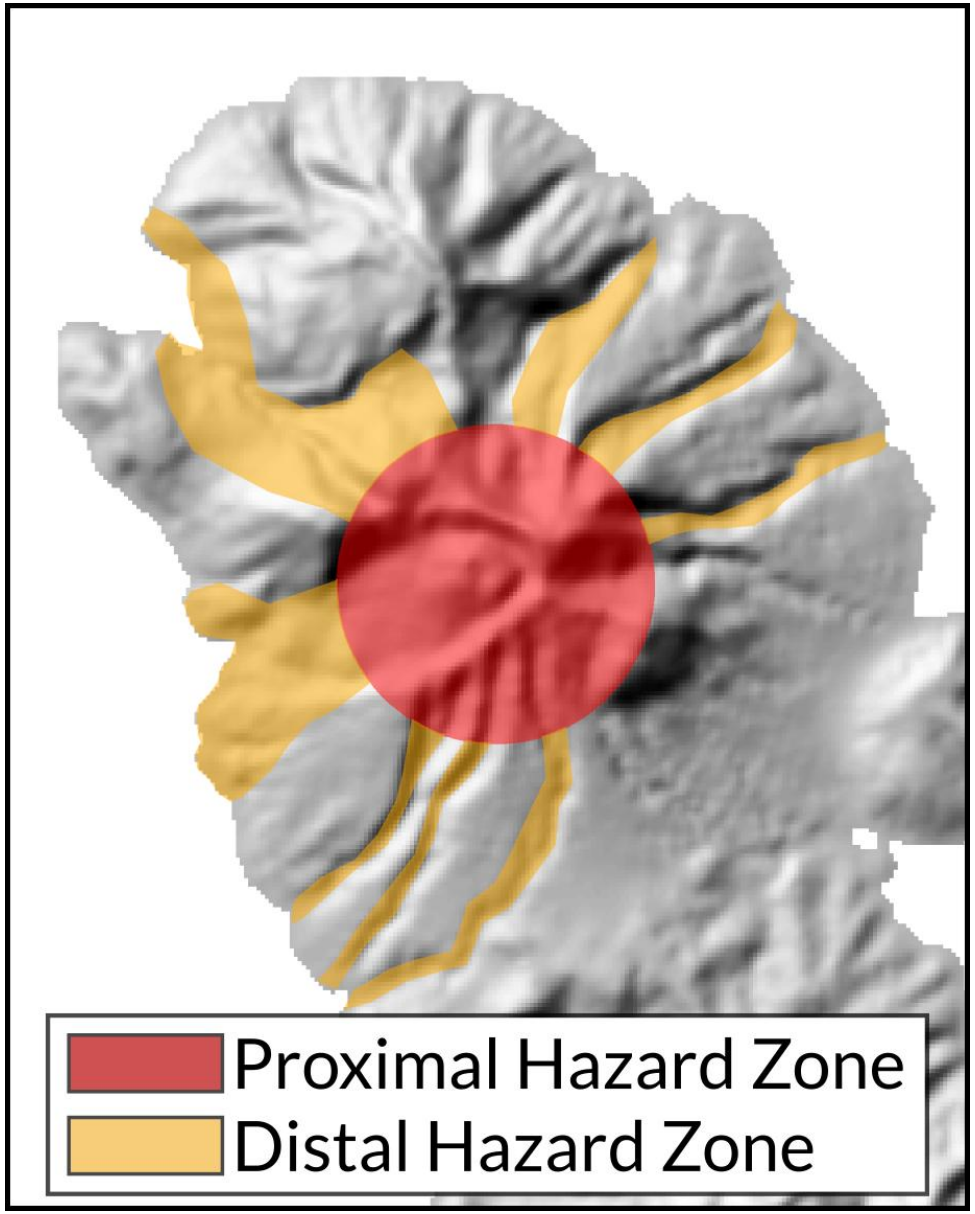
Hazard level-focused with hazard process-focused insets

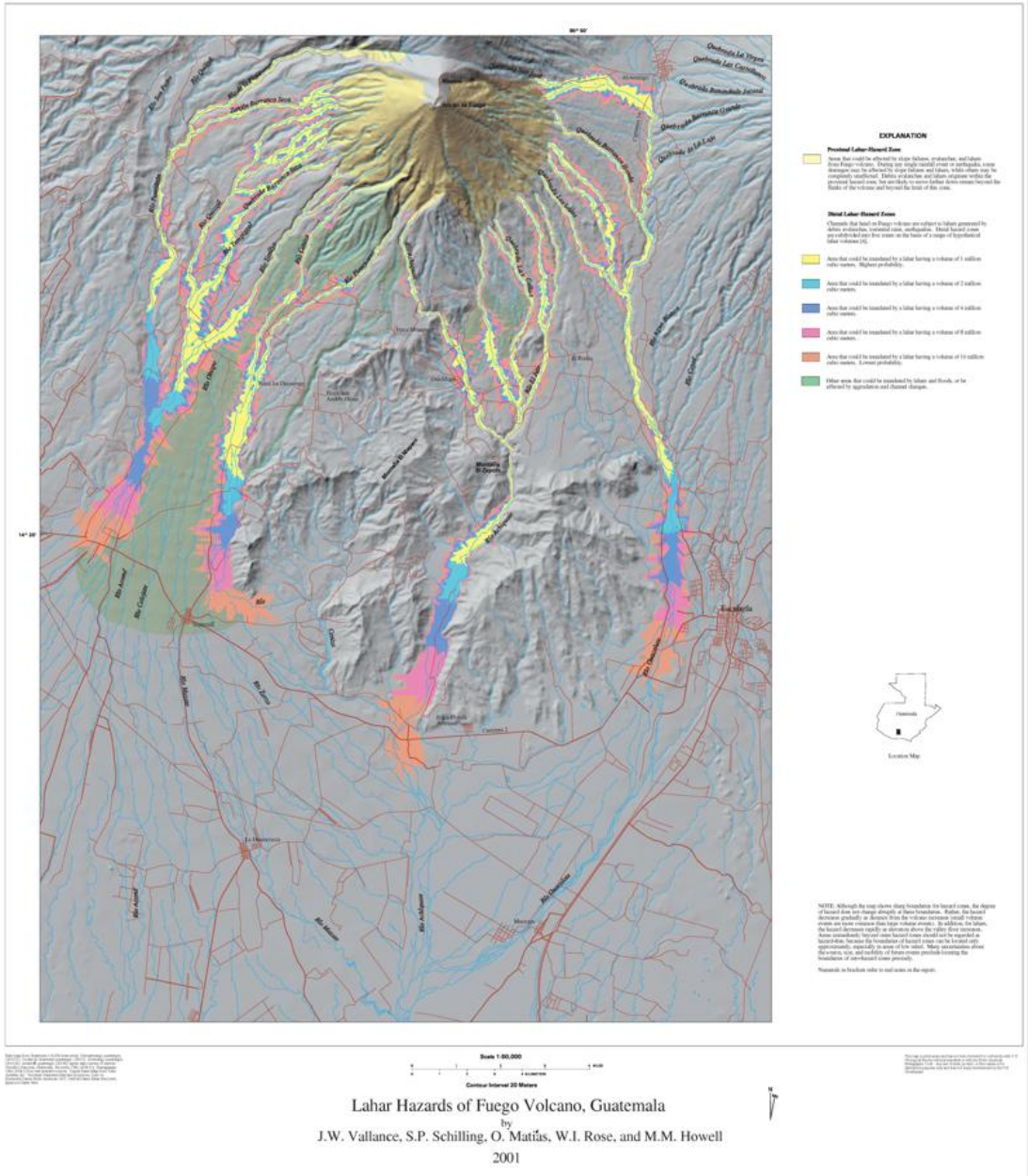




Tenerife, Spain (IGME, 2007)

Hazard process-focused (grouped by location)





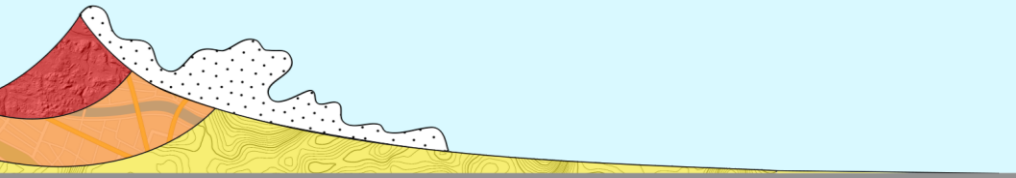
Temporal scale

Background (Long-term) Hazard Maps

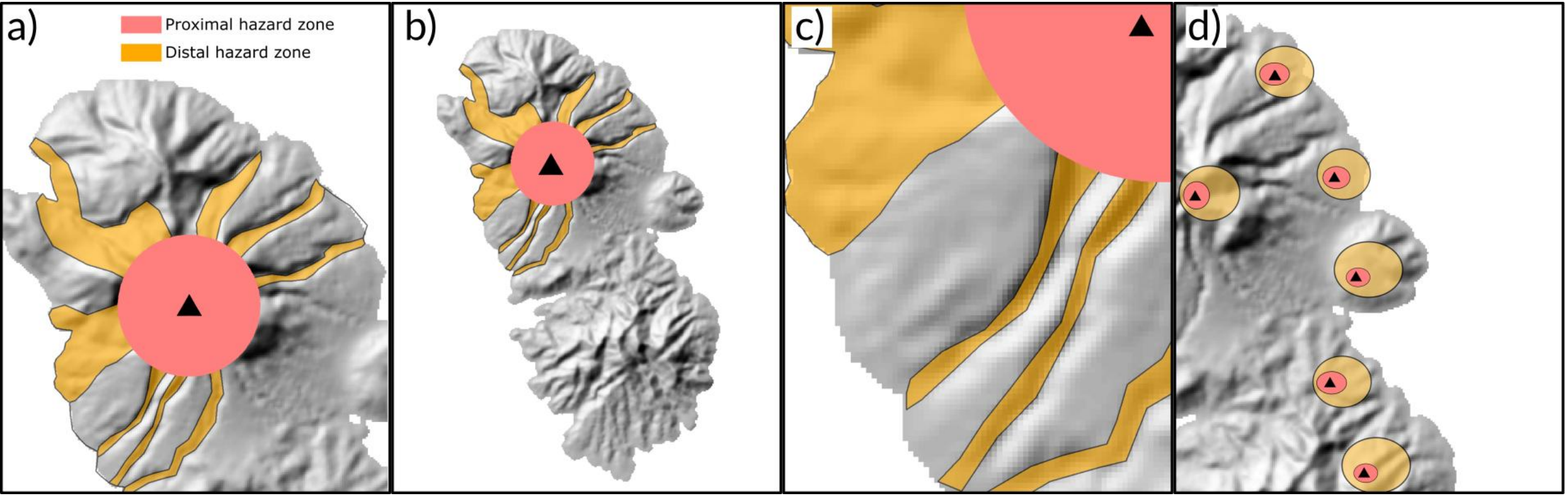
- usually created in advance of volcanic unrest
- intended to show the possible distribution of volcanic hazards over long (years to decades) time frames
- often based on a combination of methods that incorporate eruptive history, geologic records, and/or modeling
- may be based on either specific scenarios (e.g. most likely, worst-case) or on all possible activity
- low-likelihood but high-impact hazard processes may be included (e.g. lateral blasts).

Crisis (Short-term) Hazard Maps

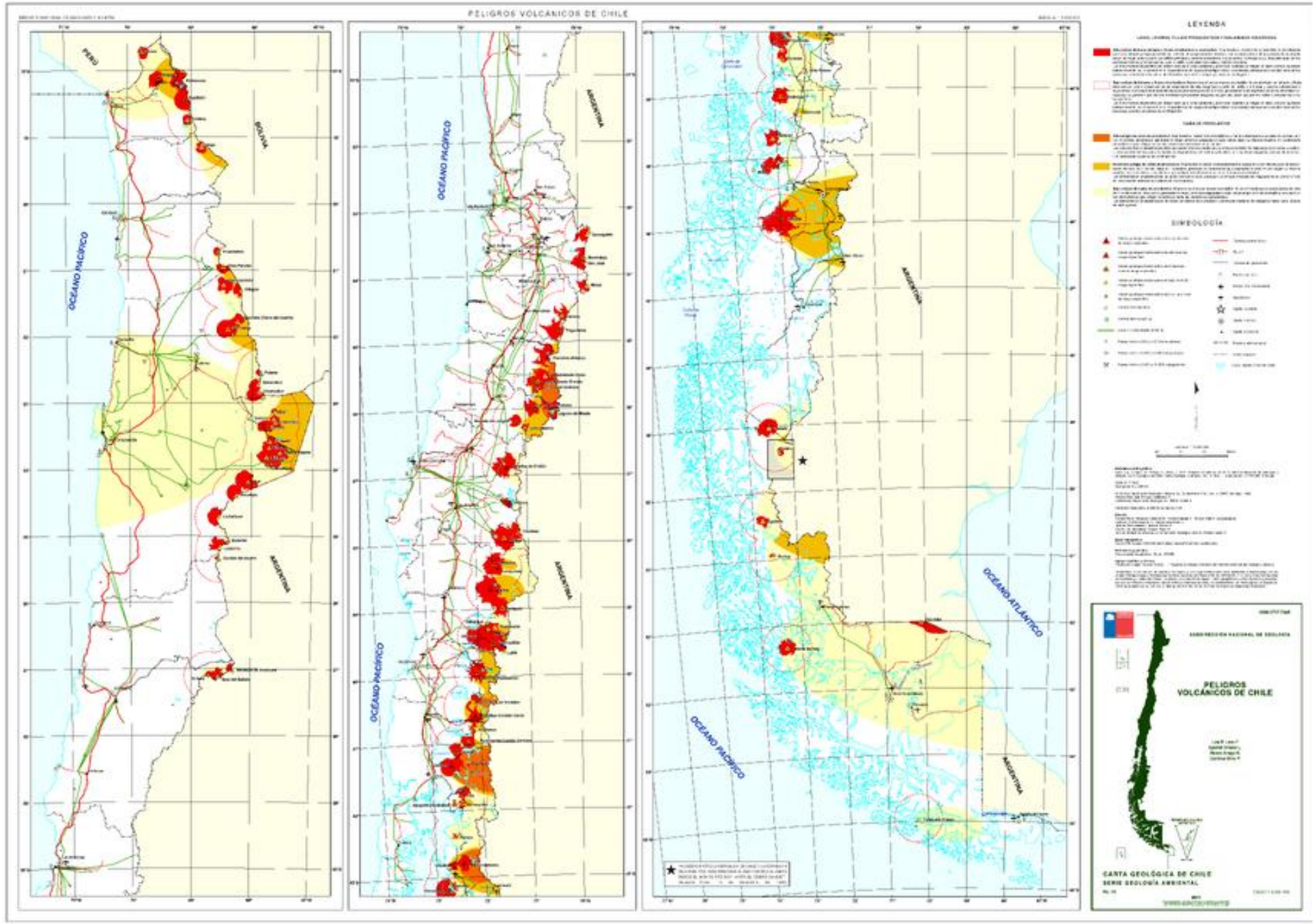
- usually created at the start of volcanic unrest or during an eruption
- usually intended for crisis-management purposes and show the likely distribution of hazards based on current conditions over short (hours to years) or very short (hours to days) time frames
- incorporate more information about the current state or specific conditions of the volcano
- commonly based on modeling, with many modeling-based crisis maps serving as forecasts with very short (hours to days) time frames



Spatial scale



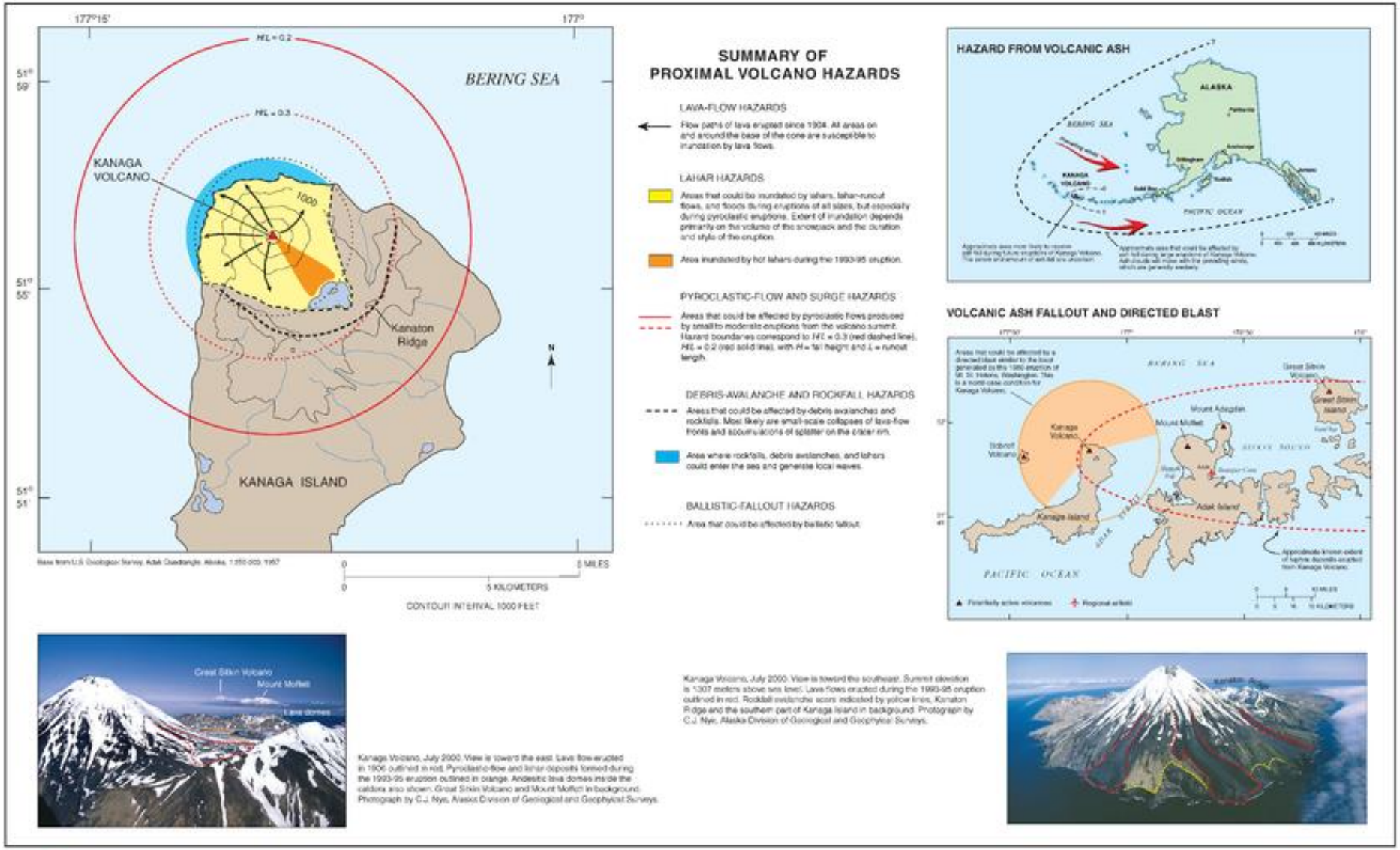
Volcano-scale | Entire island | Flank- or drainage-scale | Regional-scale



Chile (regional) (Lara et al. 2011)

U.S. DEPARTMENT OF THE INTERIOR
U.S. GEOLOGICAL SURVEY

OPEN-FILE REPORT 02–397
PLATE 1



Kanaga Volcano, July 2000. View is toward the east. Lava flow erupted in 1906 outlined in red. Pyroclastic-flow and lahar deposits formed during the 1993-95 eruption outlined in orange. Andesitic lava domes inside the caldera also shown. Great Sitkin Volcano and Mount Moffett in background. Photograph by C.J. Nye, Alaska Division of Geological and Geophysical Surveys.

Kanaga Volcano, July 2000. View is toward the southeast. Summit elevation is 1307 meters above sea level. Lava flows erupted during the 1993-95 eruption outlined in red. Rockfall avalanche scars indicated by yellow lines. Kanaton Ridge and the southern part of Kanaga Island in background. Photograph by C.J. Nye, Alaska Division of Geological and Geophysical Surveys.

PRELIMINARY VOLCANO-HAZARD ASSESSMENT FOR KANAGA VOLCANO, ALASKA

by

Christopher F. Waythomas, Thomas P. Miller, and Christopher J. Nye

2002

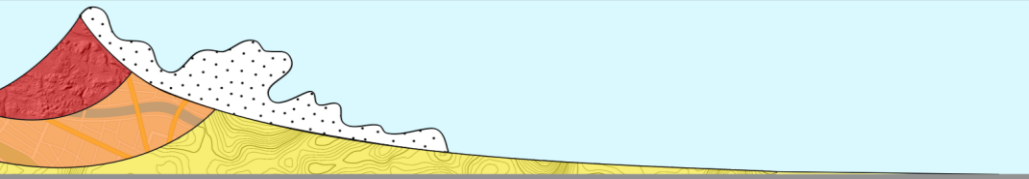
Kanaga, USA (Waythomas et al. 20021)

Purpose and Audience

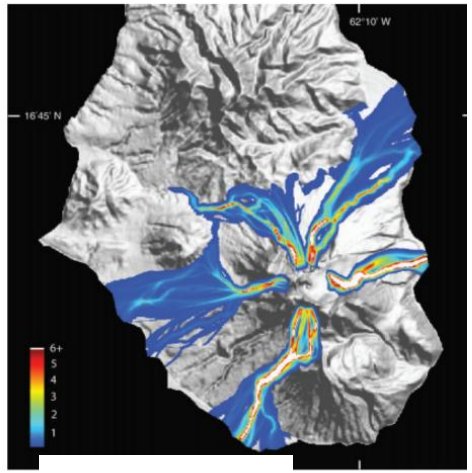
Hazard maps can also be classified according to their primary intended purpose. It can often be difficult to ascertain the map purpose unless explicitly stated or unless certain obvious design choices were made. As such, most maps with unknown purpose are designated as multi-purpose.

We welcome corrections from map-makers on intended purpose and audience designations!

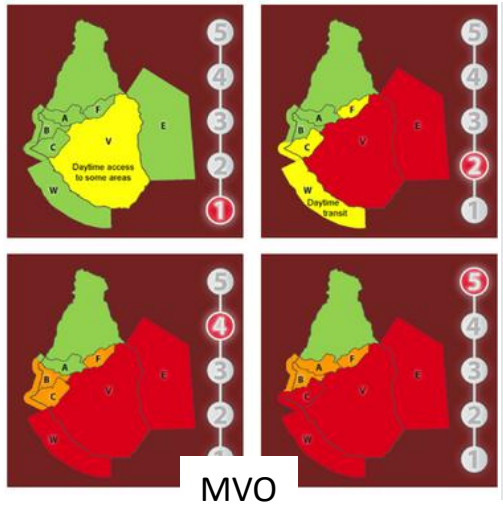
- **Hazard awareness**: Some maps are primarily aimed at educating the ***general public*** and the ***media*** about the extent and probability of volcanic hazard.
- **Land-use planning**: Some maps are primarily aimed at ***land-use planners*** for informing the citing of facilities, building restrictions, resource management, and/or infrastructure development.
- **Crisis management and response**: Some maps have the primary purpose of assisting ***civil authorities*** with management of and response to an eruption. These maps often include evacuation routes, access restrictions, actions to take, etc for the ***general public and/or tourists***
- **Situational awareness:** Some maps are intended for the situational awareness of ***federal or international governments*** or ***aid organizations***. These maps are often unofficial and derived or simplified from other hazard maps.
- **Scientific interest:** Some maps are published in academic journals for research purposes and general scientific interest; these are typically unofficial maps intended for ***scientists***.
- **Multi-purpose:** Most maps have not been specifically tailored to any one purpose, or are intended to serve all the above functions, and thus are intended for multiple purposes.



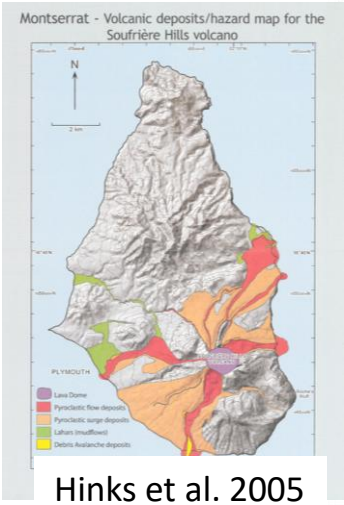
Purpose and Audience



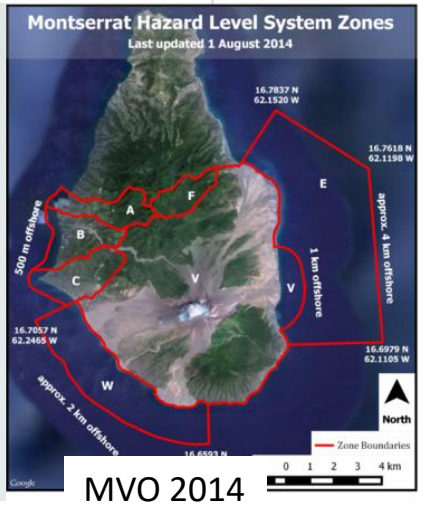
Wadge (2009)



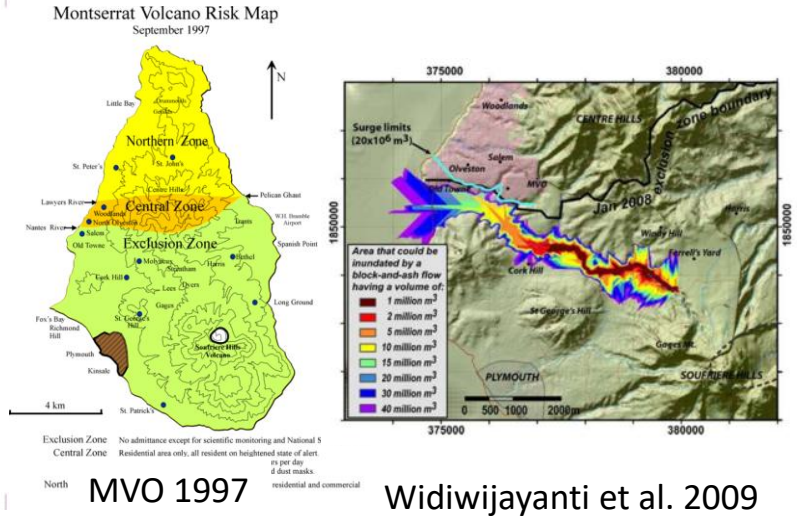
MVO



Hinks et al. 2005



MVO 2014

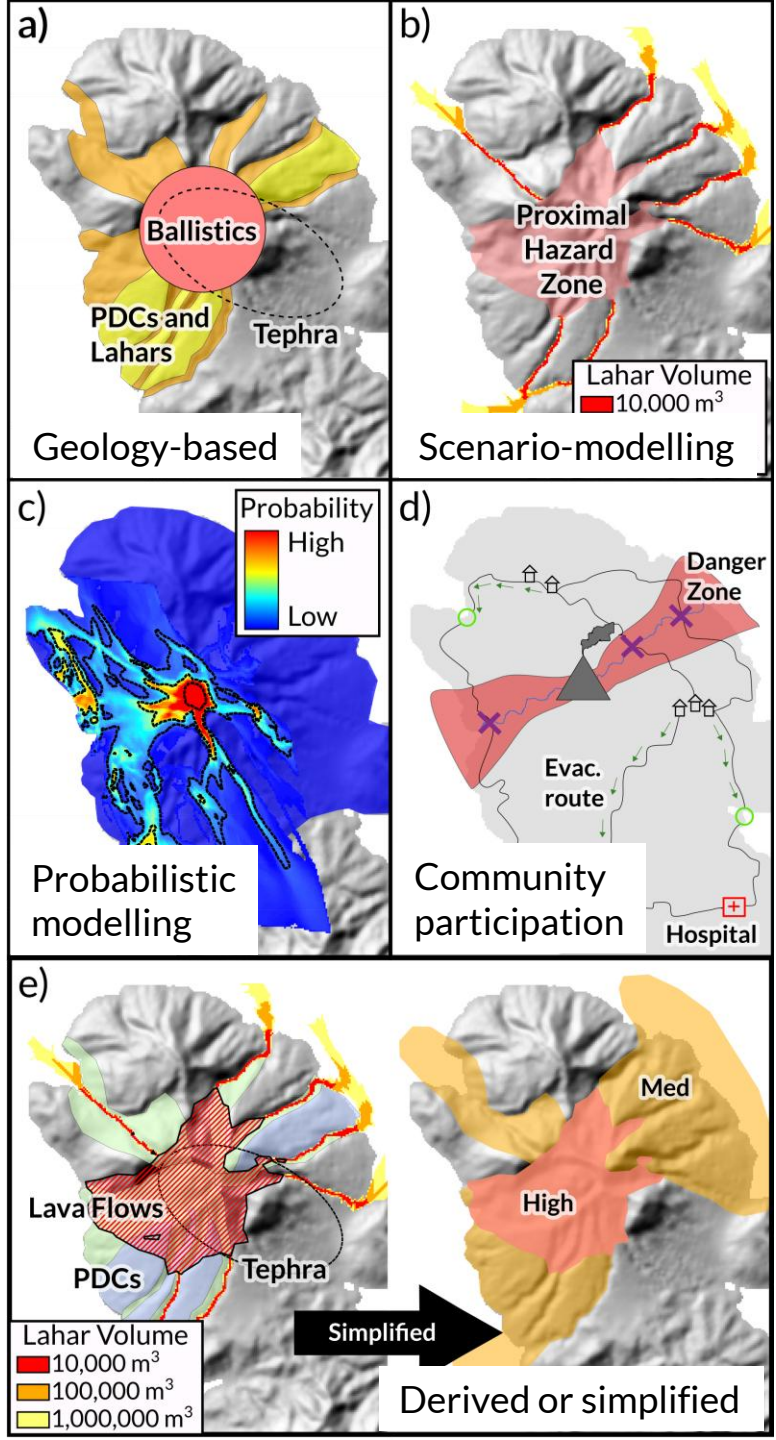


MVO 1997

Widiwijayanti et al. 2009

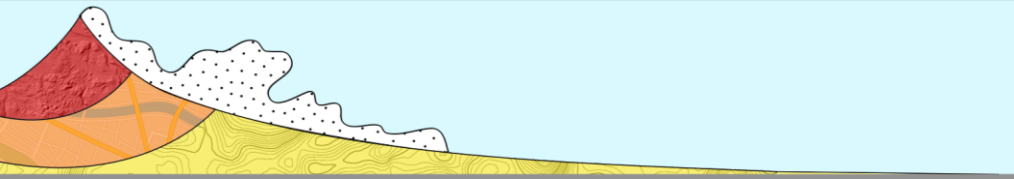
Zonation methodology

- All maps use the available geologic history of the volcano in one form or another (directly or as model inputs, for example).
- Maps often incorporate multiple methods; for example, they may use geologic history for lahar hazards and simple modeling for pyroclastic density current hazards.
- May be conditional on eruption or not.
- May consider different scenarios or not.



Scenario types

- **No specific scenario:** Many maps are based on no particular scenario or on all possible scenarios
- **Most-likely:** Most likely anticipated scenario, more common on short-term crisis maps
- **Worst-case:** worst-case or maximum-likely scenario
- **Specific past eruptions:** either at the volcano in question or those of **analog volcanoes**
- **Size**: eruptions of different sizes (e.g. VEI, small, large) or volumes of different hazardous phenomena
- **Style:** eruptions of different styles (e.g. effusive/explosive, magmatic/phreatic) or hazardous phenomena of different styles (e.g. snowmelt lahar, column-collapse PDCs)
- **Specific or current conditions**: short-term hazard maps are based on the conditions at the time the map is made
- **Source location or direction**: possible source location of future volcanic activity (e.g. likely vent- or fissure- opening locations)
- **Season or weather**: season or weather during which an eruption might occur (usually for tephra, gas, or lahar hazard)
- **Timing or duration**: Scenario related to the duration of an eruption or the timing of events
- **Topographic change**: Scenario(s) relate to possible topographic changes
- **Composition**: potential composition of a future eruption (e.g. mafic/silicic)
- **Alert level**: Scenarios related to the alert level



Models

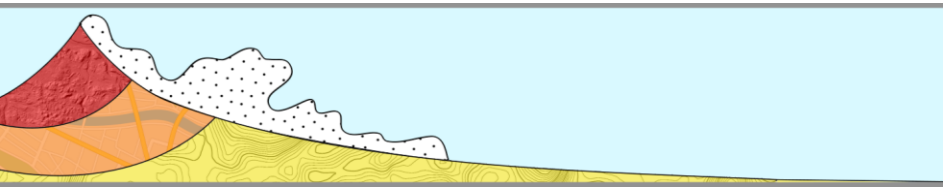
- 123 different models used by maps in the database
- Models page gives model resource links and citations
- Can search based on specific model

MAP METHODOLOGY

Method

Models Used

ASH3D
ASHFALL
Ballista
Ballistics [Bertin (2017)]
BET_VH
CALPUFF
Cellular Automata PDC model [Kelfoun et al. 1995]
Cost-distance model [Barrantes Castillo & Malavassi Rojas (2015)]
Cox process model [Diggle & Milne (1983)]
DAMBRK



Models

For maps using probabilistic or scenario-based modeling methods, there are a variety of empirical and physical models used to assess volcanic hazards. The following models have been used on at least one map in the database. The most common models used on volcanic hazard maps are the energy cone or line empirical model (14% of maps; Heim 1932; Sheridan 1979; Sheridan & Malin 1983); LAHARZ (8% of maps; Iverson et al. 1998; Schilling 1998; Schilling 2014); and Tephra2 (3% of maps; Bonadonna et al. 2005). For a full list of all maps using the listed model, click on the model name.

Ballistics

Ballista

Tsunematsu, K., Chopard, B., Falcone, J. L., & Bonadonna, C. (2014). A numerical model of ballistic transport with collisions in a volcanic setting. Computers & Geosciences, 63, p. 62-69. https://doi.org/10.1016/j.cageo.2013.10.016

Ballistics

Bertin, D. (2017). 3-D ballistic transport of ellipsoidal volcanic projectiles considering horizontal wind field and variable shape-dependent drag coefficients. Journal of Geophysical Research: Solid Earth, 122(2), p. 1126-1151. https://doi.org/10.1002/2016JB013320

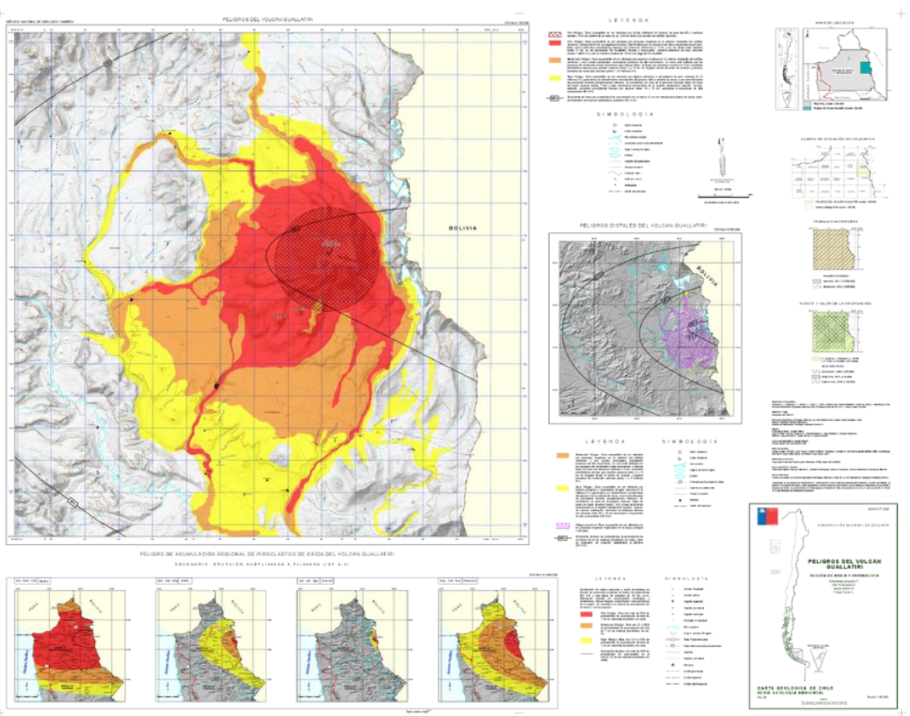
Eject!

Mastin, L.G. (2001). A simple calculator of ballistic trajectories for blocks ejected during volcanic eruptions. US Geological Survey, Open-File Report 2001-45, 13 p. https://doi.org/10.3133/ofr0145

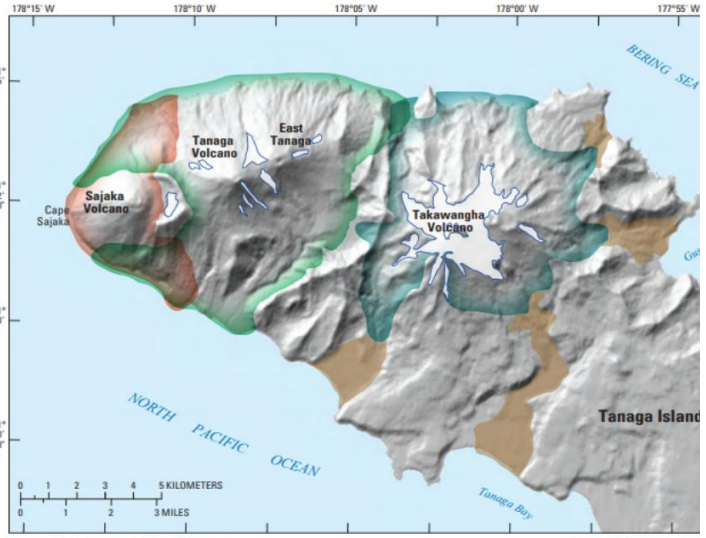
Great Balls of Fire ballistic model

Biass, S., Falcone, J. L., Bonadonna, C., Di Traglia, F., Pistolesi, M., Rosi, M., & Lestuzzi, P. (2016). Great Balls of Fire: A probabilistic approach to quantify the hazard related to ballistics—A case study at La Fossa volcano, Vulcano Island, Italy.

a)



b)



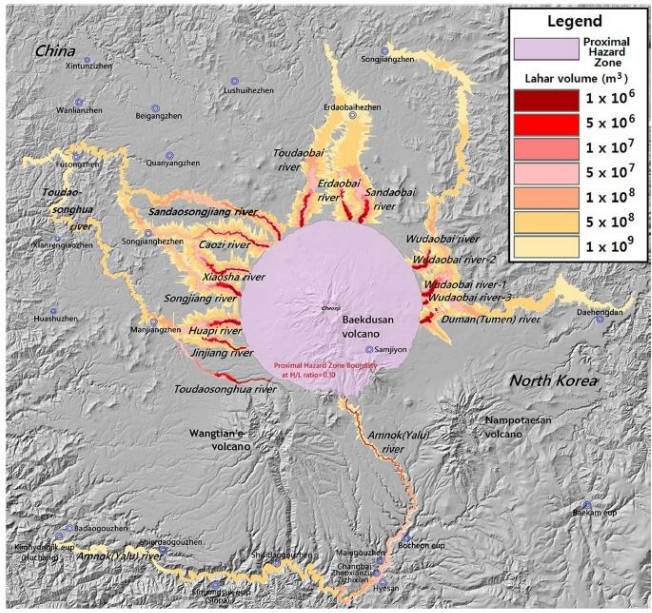
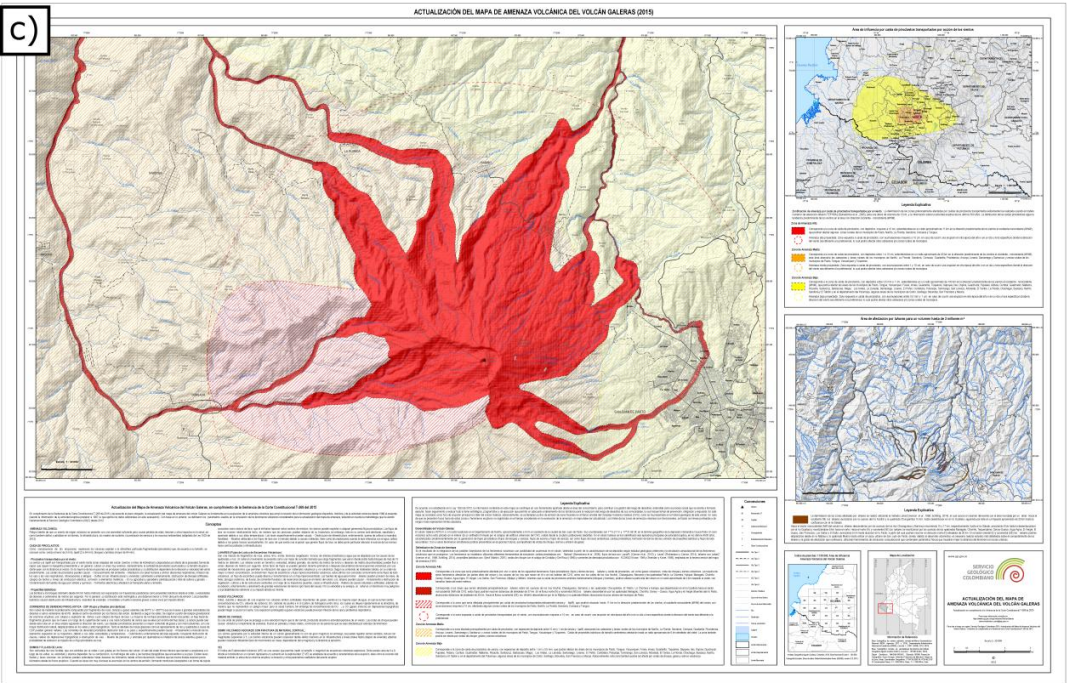


Fig. 3. Lahar-inundation hazard map constructed by applying Laharz_py program to the Mt. Baekdu volcano. The p hazard zone (H/L ratio=0.10) coloured by the violet surrounding the Mt. Baekdu is subjected to diverse hazards, includi

c)



d)

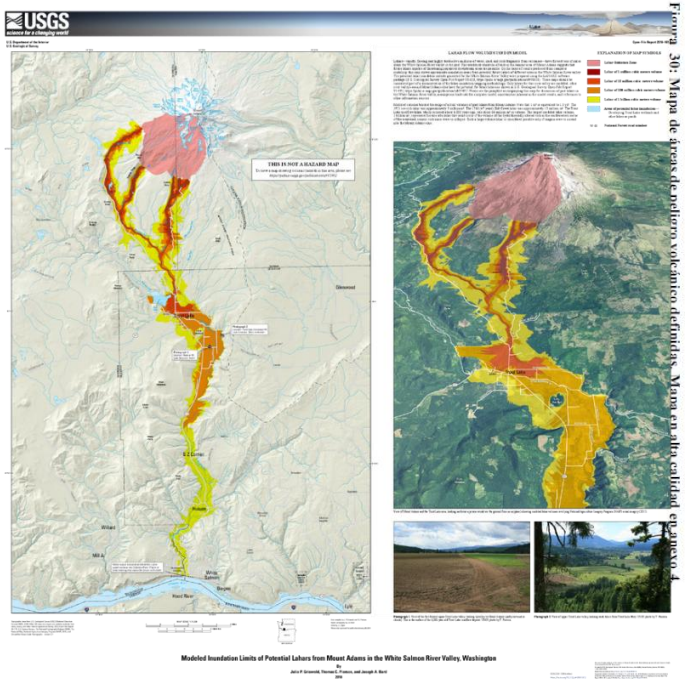
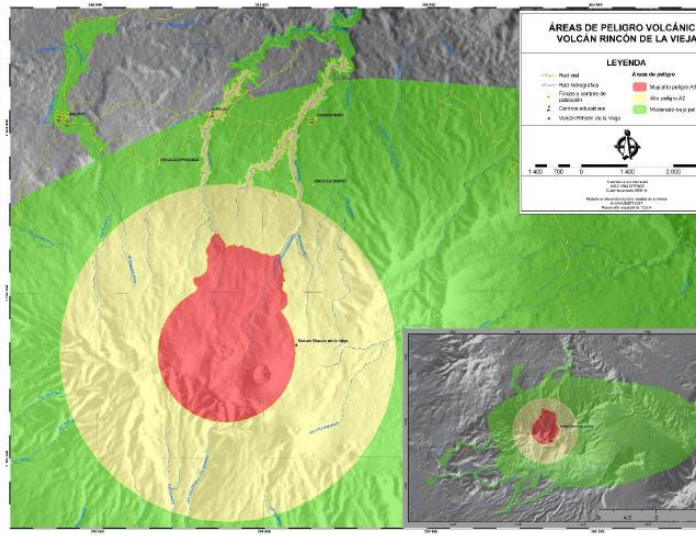


Figura 30: Mapa de áreas de peligro volcánico de lodos, Mama en alta calidad en anexo d



a)

b)

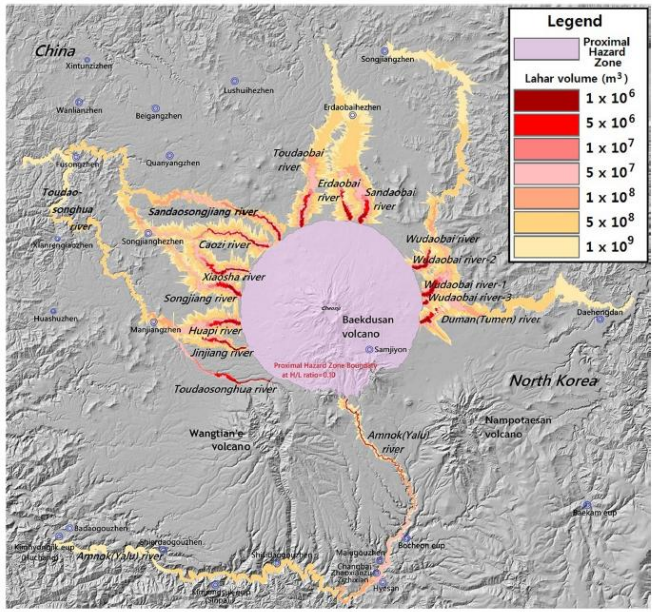
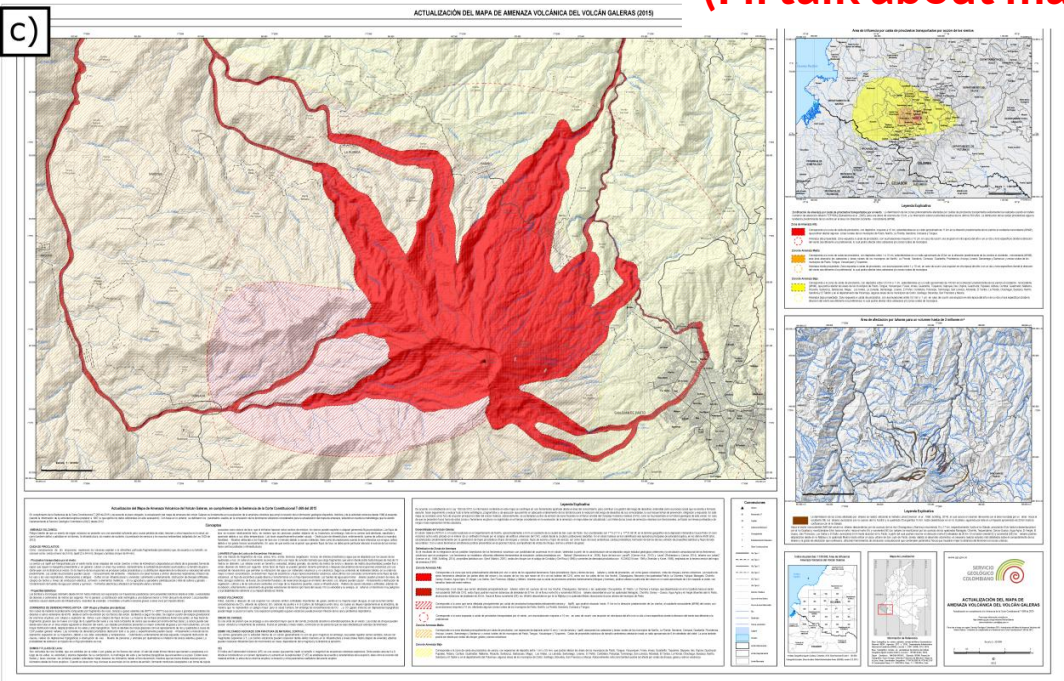


Fig. 3. Lahar-inundation hazard map constructed by applying Laharz_py program to the Mt. Baekdu volcano. The p
hazard zone (H/L ratio=0.10) coloured by the violet surrounding the Mt. Baekdu is subjected to diverse hazards, includi

The methodology need not dictate the design or "look" of the map. The final design of the map can reflect the map purpose and audience.
(I'll talk about map design another day!)

c)



d)

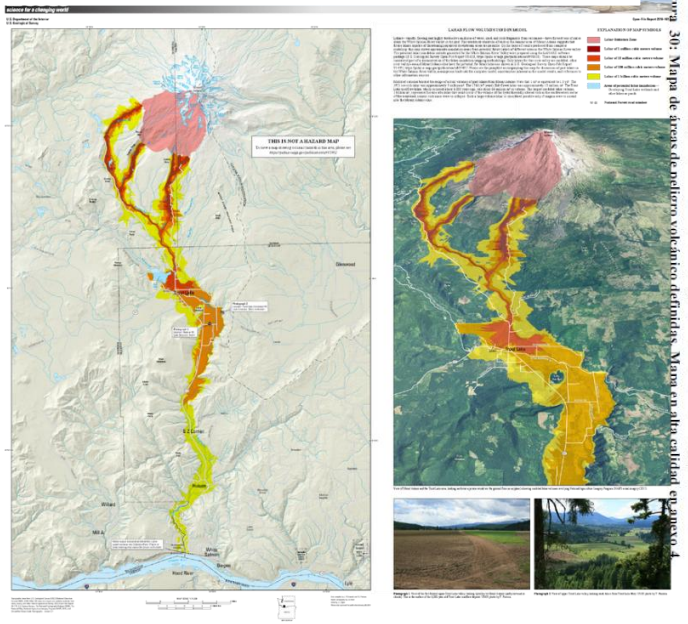
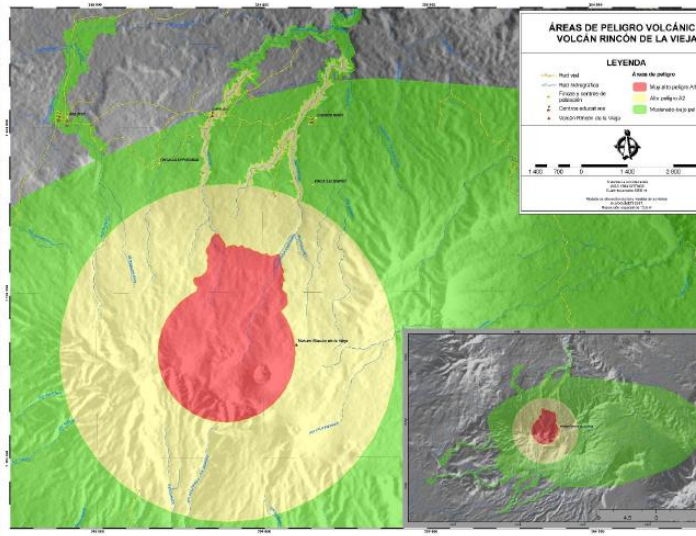


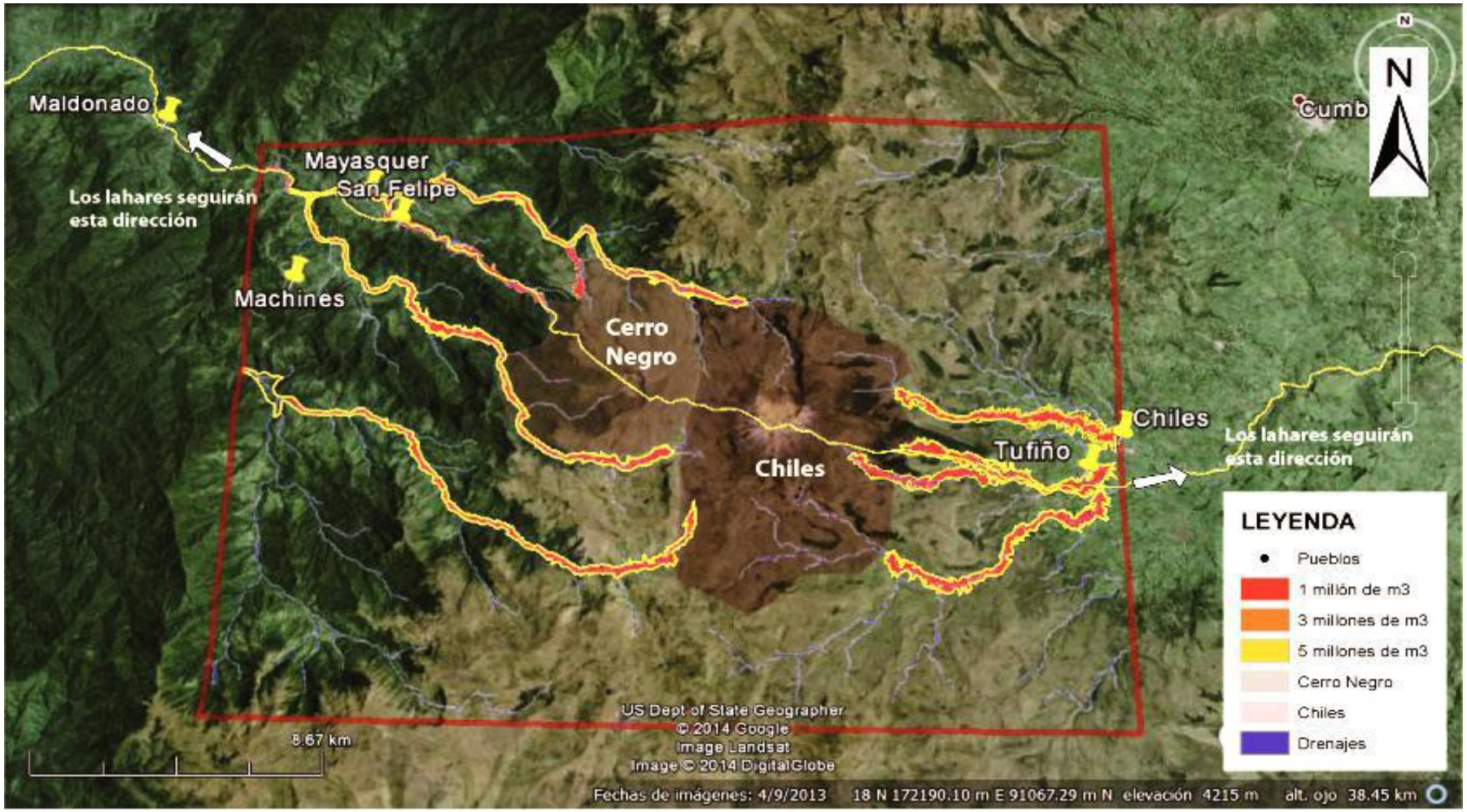
Figura 30: Mapa de áreas de peligro volcánico definidas. Mapa en alta calidad en anexo d



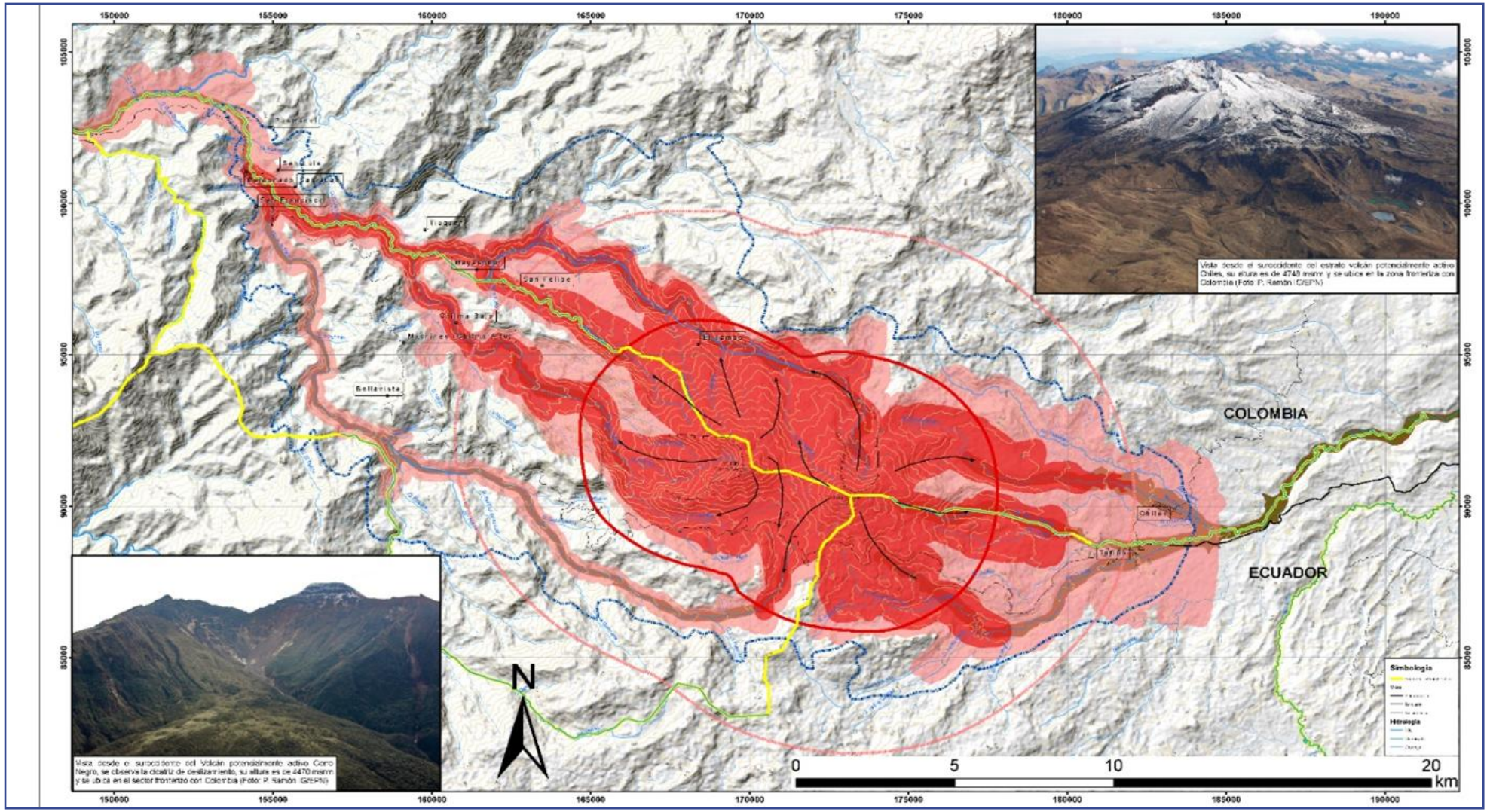
Models are just one tool!

- Models are just one tool to help you make a hazard map
- The model outputs can be modified using your geological knowledge about the volcano and about the hazards, what is known about past events, and any other additional knowledge
- **In this course we will be making maps that are conditional on eruption using scenario-based modeling (not probabilistic) AND our knowledge of geology.**

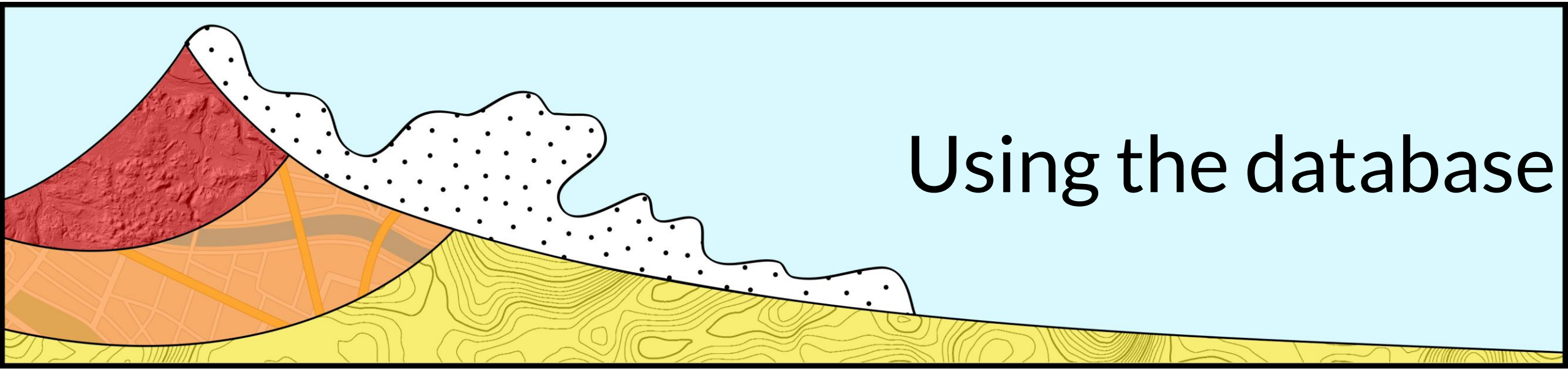




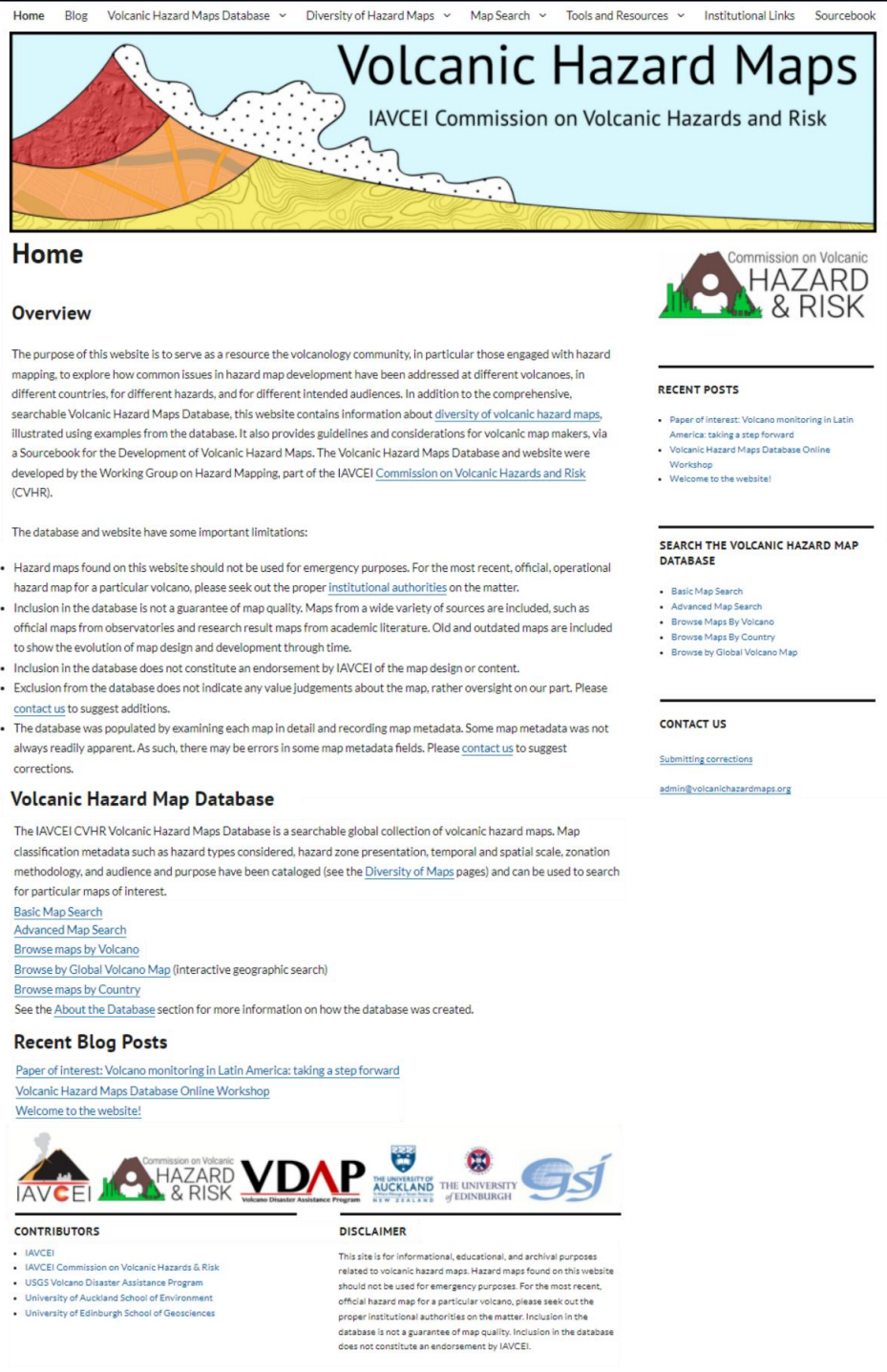
Chiles-Cerro Negro, Colombia-Ecuador border



Chiles-Cerro Negro, Colombia-Ecuador border



Using the database



Select or deselect multiple entries by holding the Ctrl key (or equivalent) while clicking.

Volcano

Any

Country

Any

Hazard zones cross international boundaries

Map Publication Year

Any to Any

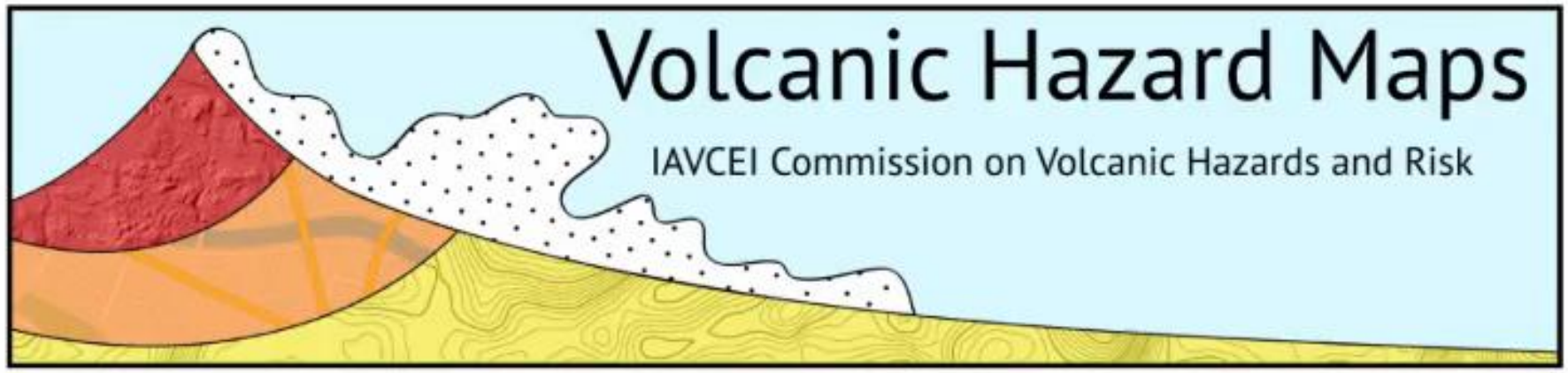
Official Most Recent

Map ID

Any

Expand filter menus by clicking the '+' symbol.

VDAP USGS



Design & Cartographic Elements

Volcanic hazard maps which show the same hazard, use the same methodology, and serve the same purpose may be visually very distinct due to the wide range of map design choices and cartographic elements included or omitted. Map design plays a huge role in how successful a map is in communicating to the user.

Jump to: color schemes, basemap types, dimensionality or view type, layout, or cartographic elements.

Color Schemes

Commission on Volcanic HAZARD & RISK

RECENT POSTS

- Paper of interest: Volcano monitoring in Latin America: taking a step forward
- Volcanic Hazard Maps Database Online

Diversity of Hazard Maps
Hazard Process Type
Hazard Zone Presentation
Temporal scale
Spatial Scale
Purpose and Audience
Language
Zonation Methodology
Models
Scenario Types
Hazard Zone Labels and Probability Definitions
Publication Format
Design & Cartographic Elements
Hazard Zone Uncertainty

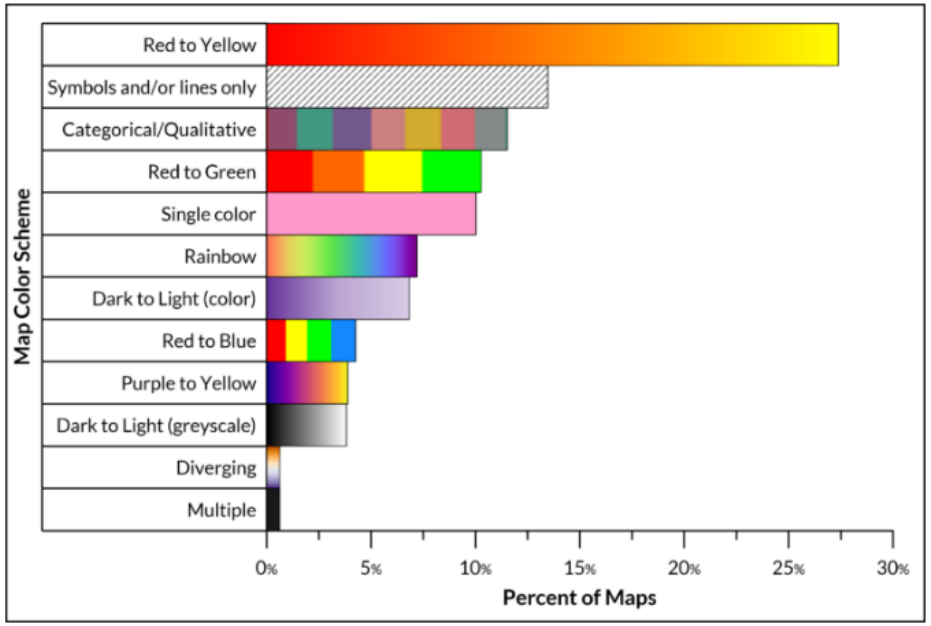


Figure 1: Percent of maps in the Volcanic Hazard Maps Database that use different color-scheme types to differentiate volcanic hazard zones . Bar colors give examples of the color-scheme in question. Current as of 2022-09-07.

Red-to-green



The logical ordering of sequential red-to-green "stoplight" color schemes (red means danger, orange means warning, yellow means caution, and green mean safe) is fairly universal among cultures (Bartels and van Beurden, 1998; Bostrom et al. 2008; Wolgalter et al., 1997) and is used globally for security warnings and traffic lights. This makes red-to-green color schemes easily understood for relative hazard levels. Roughly 10% of maps in the database use a red-to-green color scheme (**Figure 1**). About 4% of maps use a similar red-to-blue color scheme.

However, red-to-green color scales may incorrectly imply that green zones are 'safe' rather than just the lowest hazard zone (Monmonier 2018). Additionally, red-to-green color schemes can pose issues for the color-blind because the highest

Basemap Type

Volcanic hazard maps are displayed on a variety of different kinds of basemaps (**Figure 2**). The choice of basemap depends upon the target audience, with some basemaps more easily understood by the general public, for example, than others. Studies have shown (e.g. Haynes et al. 2007) that different basemaps and view types (2D planimetric vs. 3D oblique) are better for different map tasks, such as orienting (finding north), locating (identifying one's neighborhood, for example), map skills (identifying roads, hills, rivers, etc.), and understanding hazard information. Maps often have multiple types of basemaps using overlays; for example, a hillshade basemap may also have contours lines and streets overlain on it. Therefore, the percentages presented in this section sum to more than 100%.

Digital Elevation Model (DEM), Hillshade, or TIN

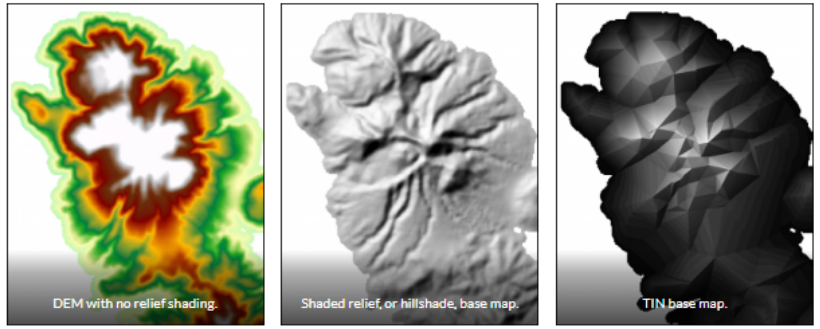


Figure 2: Cartoon examples of a DEM base map, a shaded relief or hillshade base map, and TIN base map.

Digital Elevation Models (DEMs), shaded relief or hillshade maps, and Triangulated Irregular Networks (TINs) are all digital representations of topography (**Figure 2**).

Hillshade, or shaded relief, maps are based on DEMs but simulate shadows to emphasize terrain relief and thus mimic the look of real topography shaded by the sun. Hillshades are the most common basemap used by volcanic hazard maps in the database (44%). Studies have shown that hillshade maps are much more intuitive for topography interpretation than contour maps (Haugerud & Greenberg 1998; Konnelly 2002).

Hillshades can be created from a DEM using a GIS and specifying the azimuth and elevation of the simulated sun. Default settings typically place the sun at the upper-left corner of the map. Placement of the simulated light source at the lower-right corner of the map can make the topography appear inverted due to perceptual illusions, and should be avoided (Saraf et al. 1996; Bernabé-Poveda & Çöltekin 2014).

Google Earth Placemark with Features Cite Volcano Profile

Latest Activity Reports | Weekly Reports | Bulletin Reports | Synonyms & Subfeatures | General Information

Eruptive History | Deformation History | Emission History | Photo Gallery | Map Holdings | Sample Collection | External Sites

External Sites

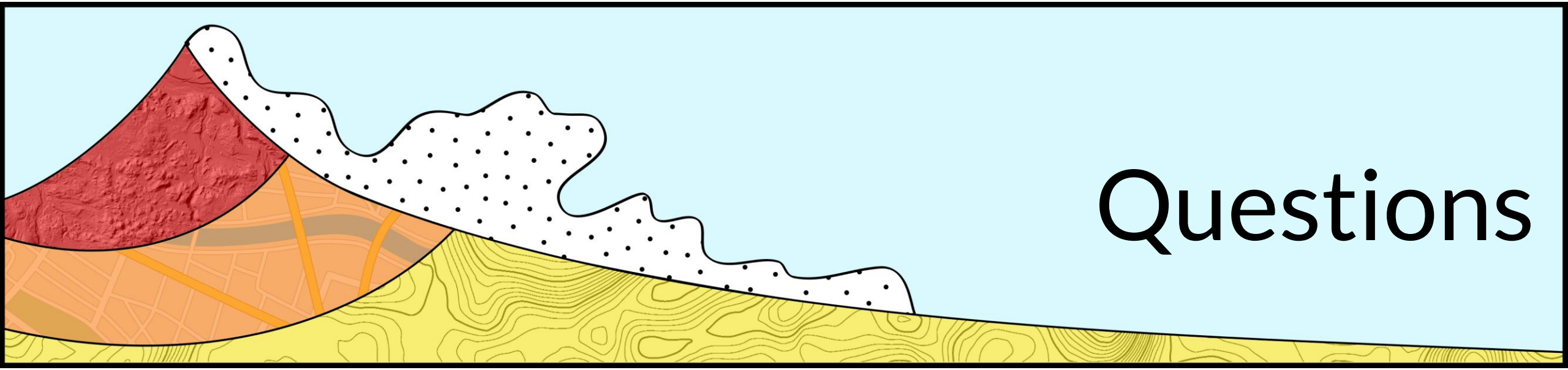
Volcanic Hazard Maps

IAVCEI Commission on Volcanci Hazards and Risk has a Volcanic Hazard Maps database designed to serve as a resource for hazard mappers (or other interested parties) to explore how common issues in hazard map development have been addressed at different volcanoes, in different countries, for different hazards, and for different intended audiences. In addition to the comprehensive, searchable Volcanic Hazard Maps Database, this website contains information about diversity of volcanic hazard maps, illustrated using examples from the database. *This site is for educational purposes related to volcanic hazard maps. Hazard maps found on this website should not be used for emergency purposes. For the most recent, official hazard map for a particular volcano, please seek out the proper institutional authorities on the matter.*

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- The hazard map database was populated by examining each map in detail and recording map metadata. Some map metadata, such as ***map purpose, audience, and zonation methodology*** was not always readily apparent. In these cases, best judgment was used to classify the map. As such, there may be errors in some map metadata fields, which can be corrected if brought to our attention at admin@volcanichazardmaps.org.
- If writing to make a correction, please include your name, institution, the name and url of the map needing correction, and a description of the problem.
- To suggest an addition to the database, please include your name, institution, and as much information about the suggested map as possible, including a url, if possible.
- Map thumbnail images are displayed only for maps for which we have obtained copyright. To provide copyright permissions, please complete this form and email it to admin@volcanichazardmaps.org.





Questions

Website disclaimers

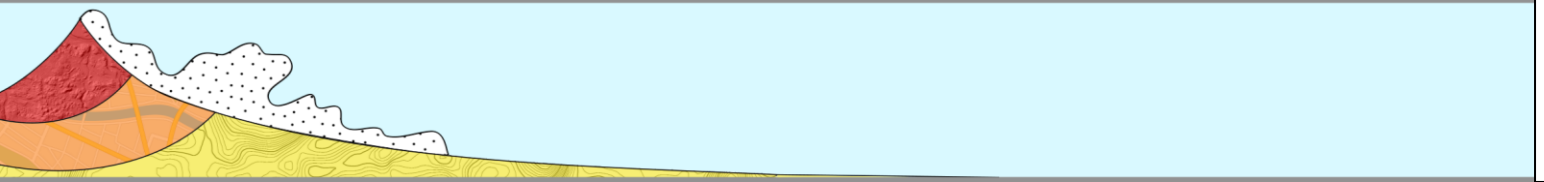
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This map was produced by an official agency, but may not be the most recent, most complete, main operational map. For the current operational hazard map, please consult the proper institutional authorities.

This map was produced by an official agency of a foreign government, and is NOT the most recent, most complete, main operational map. For the current operational hazard map, please consult the proper institutional authorities.

This is an unofficial hazard map intended for scientific research only. This map should not be used for any purpose other than scientific interest; it is presented here to serve as an example of the map-making methodology. For the current operational hazard map, please consult the proper institutional authorities.

This is an unofficial hazard map created for the situational awareness of an international agency. This map should not be used for any purpose; it is presented here to serve as an example of this map type. For the current operational hazard map, please consult the proper institutional authorities.



DISCLAIMER

This site is for educational purposes related to volcanic hazard maps. Inclusion in the database is not a guarantee of map quality. Hazard maps found on this website should not be used for emergency purposes. For the most recent, official hazard map for a particular volcano, please seek out the proper institutional authorities on the matter. Inclusion in the database does not constitute an endorsement by IAVCEI.

Institutional Links

Argentina

Servicio Geológico Minero Argentino (SEGEMAR)

Canada

Natural Resources Canada (NRC)

Chile

Servicio Nacional de Geología y Minería (SERNAGEOMIN) : Mapas de amenaza o peligro volcánico [Volcanic hazard or danger maps]

Servicio Nacional de Geología y Minería (SERNAGEOMIN) : Red Nacional de Vigilancia Volcánica [National Volcanic Surveillance Network]

Servicio Nacional de Geología y Minería (SERNAGEOMIN) : Catálogo Virtual E-Maps [Virtual E-Maps Catalog]

Colombia

Servicio Geológico Colombiano (SGC) : Mapas de Amenaza Volcánica [Volcanic Hazard Maps]

Costa Rica

Observatorio Vulcanológico y Sismológico de Costa Rica (OVSICORI)

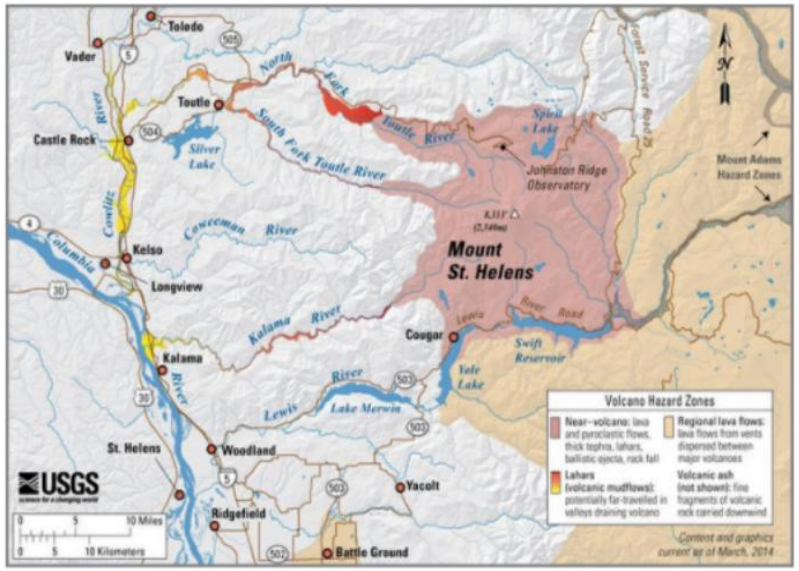
DR Congo

Mount St. Helens, Washington simplified hazards map showing potential impact area for ground-based hazards during a volcanic event.

St. Helens, United States (2014)

U.S. Geological Survey (USGS). (2014). Mount St. Helens, Washington simplified hazards map showing potential impact area for ground-based hazards during a volcanic event. US Geological Survey simplified hazard map.

This map was produced by an official agency, but may not be the most recent, most complete, main operational map. For the current operational hazard map, please consult the proper institutional authorities.



Click on image for full size map.

Map Data

Hazard Processes	Ballistics; Lahars; Lava flows; PDCs; Rockfalls; Tephra fall
Hazard Presentation	Hazard process-focused (grouped by location): Hazard processes are grouped into proximal/distal or other location-based zones on a main map panel
Temporal Scale	Background, or long-term, map
Spatial Scale	Volcano and surrounding area
Publication Format	Map on website
Zonation Method(s)	Derived/simplified from another map
Zonation Model(s)	Map on website
Scenario Type(s)	Location- or direction-based
Hazard Zone Label(s)	Hazard name; Location or source name
Probability Definition(s)	Qualitative relative probability (e.g. high-medium-low)
Purpose	Hazard awareness: Intended for education the audience about the extent and probability of volcanic hazard
Audience	General public and/or media
Language(s)	English
Basemap(s)	Hillshade; Street map
Dimensionality	Planimetric (map) view
Color Scheme	Red to Yellow

Future development

- **Database additions**:
  - **Risk maps**: While some maps with risk information exist in the database, we have not specifically sought out risk maps, this is a possible future addition.
  - **Hazard assessments**: Maps from hazard assessments (pocket inserts and figures) are included, we haven't collected information about the contents of the hazard assessments themselves
  - **New maps**
- **Map prototyping, map comparison tools?**
- **Data download tool**
- **User corrections/additions**: we'd like to add a correction and map addition form so users could easily submit changes. For now, use the "Contact Us" page to email me!

